ICCV OCT 19-23, 2025 HONOLULU HAWAII

Foundation Models in Visual Anomaly Detection: Advances, Challenges, and Applications

Organizers: Jiawen Zhu, Peng Wu, Olga Fink, Chengjie Wang, Guansong Pang

Guest Speakers: Ismail Nejjar, Jinlong Peng

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Foundation Models in Video Anomaly Detection

Peng Wu

School of Compute Science
Northwestern Polytenical University



西北工業大學
NORTHWESTERN POLYTECHNICAL UNIVERSITY

Outline

1. Research Background
2. VLM-based Video Anomaly Detection
3. VLM-based Video Anomaly Retrieval
4. VLM-based Video Anomaly Caption
5. VLM-based Open Anomaly Detection

1 Research Background

Video Anomaly Understanding (VAU) aims to automatically identify abnormal behaviors in surveillance and related scenarios. With the surge of large-scale video data, achieving efficient and interpretable anomaly understanding has become a key challenge.



Intelligent surveillance



Intelligent transportation



Intelligent content review

1 Research Background

VLMs provide a new paradigm for building more generalizable and interpretable VAU systems.

leveraging the powerful representation ability, modality alignment, semantic reasoning, and zero-shot generalization capabilities of VLMs.

Representation: Extracts high-quality cross-modal feature embeddings.

Alignment: Achieves deep alignment between visual content and linguistic descriptions.

Reasoning: Generates interpretable explanations based on visual cues and prior knowledge.

Generalization: Recognizes unseen categories and adapts to open-world scenarios.

VAD--Detection

Locating video events that deviate from the normal patterns



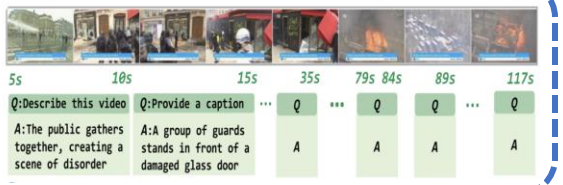
VAR--Retrieval

Locating related events in large-scale videos based on text queries



VAC--Caption

providing semantic language explanations for detection results.



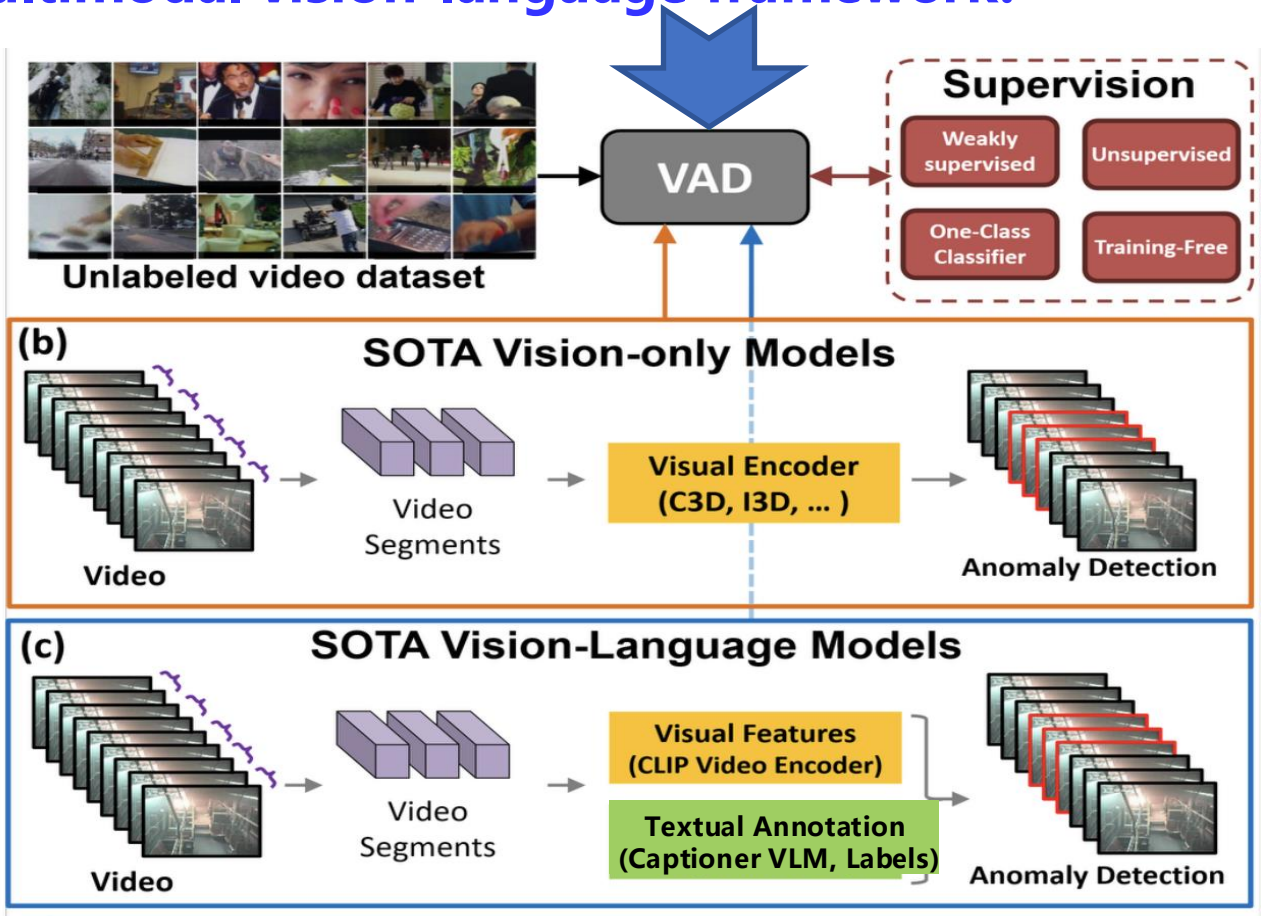
OVAD--Open

identifying unseen anomaly types absent during training.



2 Overview of VLM-based VAD

Current VAD methods mainly follow a "feature extraction + video frame classification" paradigm. With the rise of VLMs, this paradigm is evolving from purely visual classification to a multimodal vision-language framework.

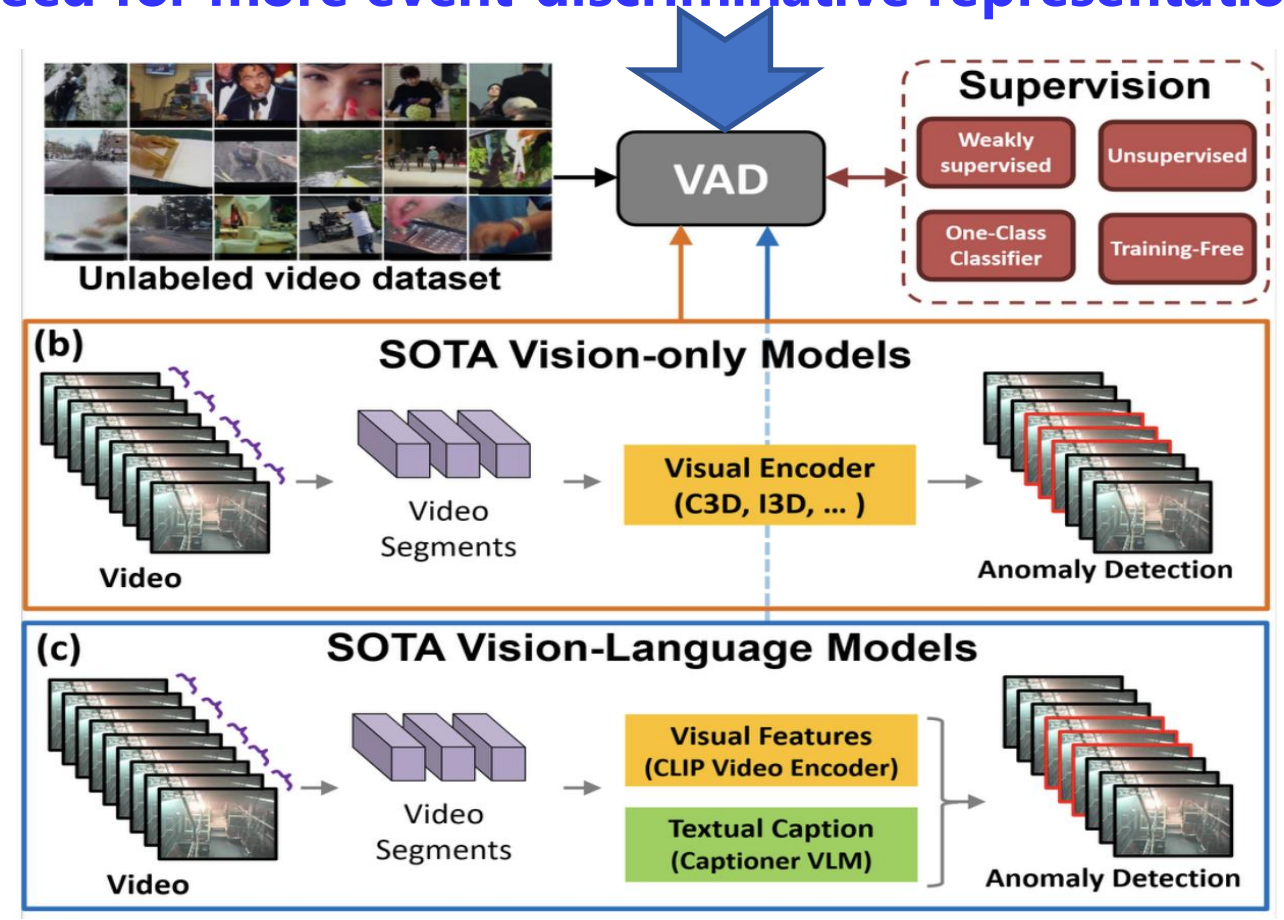


2 Overview of VLM-based VAD

Representation in the Early Exploration of VLMs

Proble m

Current VAD methods use action recognition backbones like C3D and I3D for visual encoding, but these models are ill-suited for VAD due to the domain gap and the need for more event-discriminative representations.

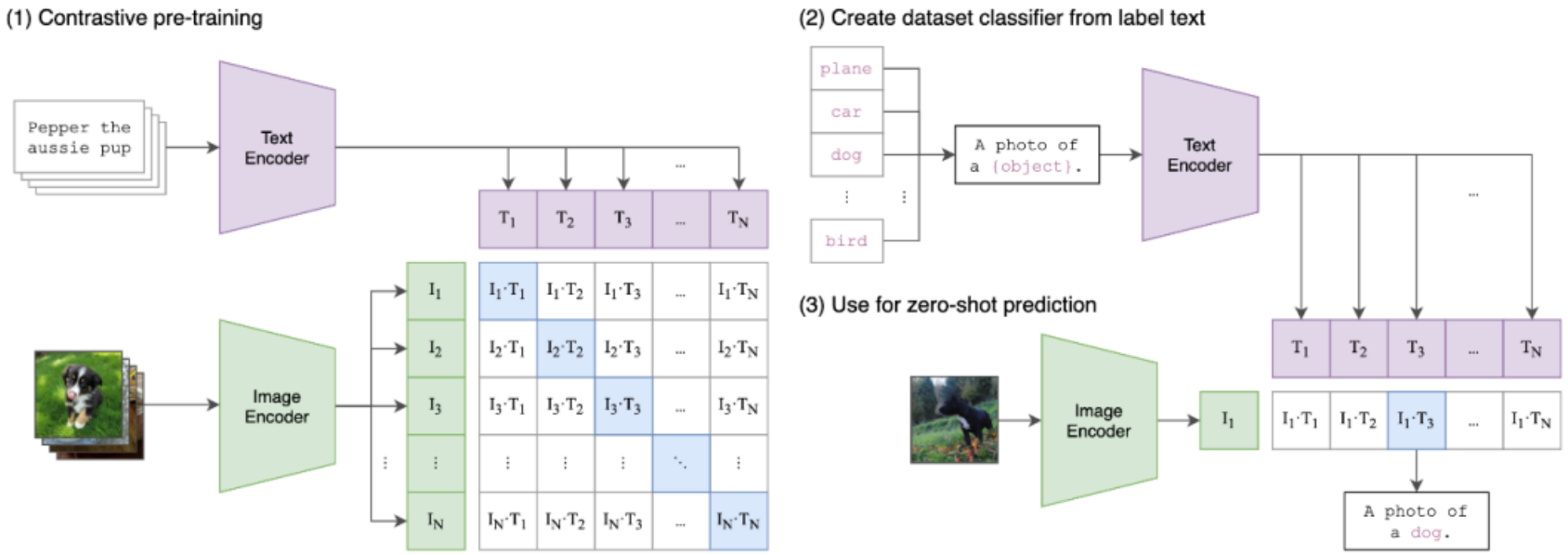


2 Overview of VLM-based VAD

Representation in the Early Exploration of VLMs

Thought

For anomaly detection, VLM features offer stronger separability and clearer semantic information than traditional extractors like C3D or I3D.

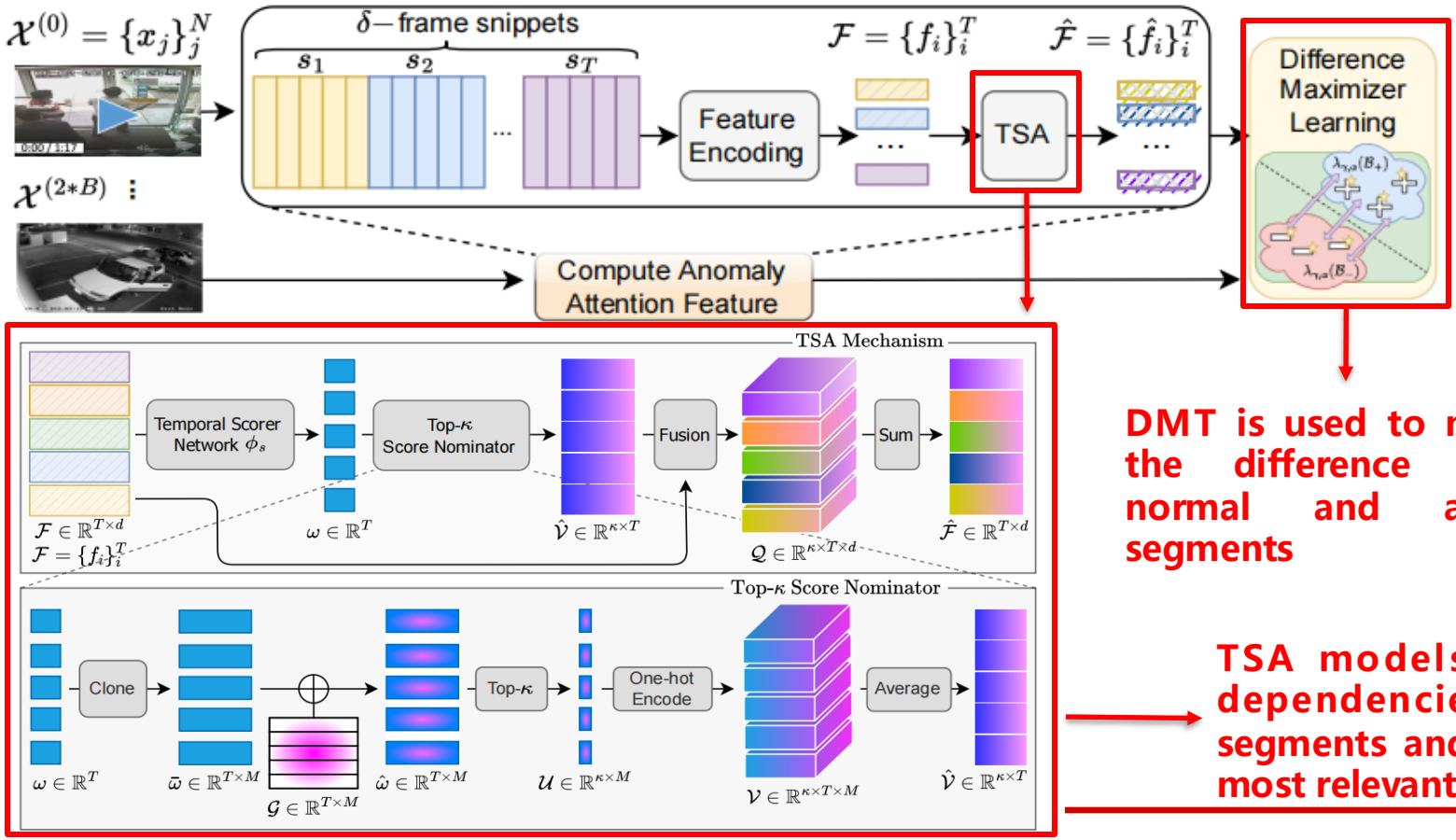


2 Overview of VLM-based VAD

Representation in the Early Exploration of VLMs

Method

CLIP-TSA effectively detects video anomalies by combining CLIP-based visual features with a Temporal Self-Attention (TSA) and a Difference Maximization Trainer (DMT).



DMT is used to maximize the difference between normal and abnormal segments

TSA models temporal dependencies between segments and selects the most relevant ones.

Joo H K, Vo K, Yamazaki K, et al. Clip-tsa: Clip-assisted temporal self-attention for weakly-supervised video anomaly detection. **ICIP**, 2023.

2 Overview of VLM-based VAD

Representation in the Early Exploration of VLMs

Result

ACLIP-TSA achieves state-of-the-art performance on the UCF, XD-Violence, and ShanghaiTech datasets.

Sup.	Method	Venue	Feature	AUC@ROC ↑
Un-	Lu et al. [31]	ICCV'13	C3D	65.51
	Hasan [32]	CVPR'16	-	50.60
	BODS [33]	ICCV'19	I3D	68.26
	GODS [33]	ICCV'19	I3D	70.46
	GCL [34]	CVPR'22	ResNext	71.04
Fully-	Liu & Ma [16]	MM'19	NLN	82.0
Weakly-	GCN [21]	CVPR'19	TSN	82.12
	GCL_{WS} [34]	CVPR'21	ResNext	79.84
	Purwanto et al. [35]	ICCV'21	TRN	85.00
	Thakare et al. [36]	ExpSys'22	C3D+I3D	84.48
	Sultani et al. [18]	CVPR'18	C3D	75.41
	Zhang et al. [37]	ICIP'19		78.70
	GCN [21]	CVPR'19		81.08
	CLAWS [19]	ECCV'20		83.03
	RTFM [38]	ICCV'21		83.28
	Sultani et al. [18]	CVPR'18	I3D	77.92
	Wu et al. [39]	ECCV'20		82.44
	DAM [40]	AVSS'21		82.67
	RTFM [38]	ICCV'21		84.30
	Wu & Liu [41]	TIP'21		84.89
	BN-SVP [42]	CVPR'22		83.39
	Ours: CLIP-TSA		CLIP	**87.58**

Sup.	Modality	Method	Venue	Feature	AUC@PR ↑
Un-	–	OCSVM [43]	NeurIPS'00	–	27.25
		Hasan et al. [32]	CVPR'16	–	30.77
Weakly-	Vision & Audio	Wu et al. [39]	ECCV'20	I3D(V) + VGGish(A)	78.64
		Wu & Liu [41]	TIP'21	I3D(V) + VGGish(A)	75.90
		Pang et al. [44]	ICASSP'21	I3D(V) + VGGish(A)	81.69
		MACIL-SD [45]	MM'22	I3D(V) + VGGish(A)	83.40
		DDL [46]	ICCECE'22	I3D(V) + VGGish(A)	**83.54**
	Vision	Sultani et al. [18]	CVPR'18	C3D(V)	73.20
		RTFM [38]	ICCV'21	C3D(V)	75.89
		RTFM [38]	ICCV'21	I3D(V)	77.81
		Ours: CLIP-TSA		CLIP(V)	82.19

Sup.	Method	Venue	Feature	AUC@ROC ↑
Un-	Hasan et al. [32]	CVPR'16	-	60.85
	Gao et al. [48]	ICCV'19	-	71.20
	Yu et al. [49]	MM'20	-	74.48
	GCL_{PT} [34]	CVPR'21	ResNext	78.93
Weakly-	GCN [21]	CVPR'19	TSN	84.44
	GCL_{WS} [34]	CVPR'21	ResNext	86.21
	Purwanto et al. [35]	ICCV'21	TRN	96.85
	GCN [21]	CVPR'19	C3D	76.44
	Zhang et al. [37]	ICIP'19		82.50
	CLAWS [19]	ECCV'20		89.67
	RTFM [38]	ICCV'21		91.57
	BN-SVP [42]	CVPR'22		96.00
	Sultani et al. [18]	CVPR'18	I3D	85.33
	AR-Net [50]	ICME'20		91.24
	DAM [40]	AVSS'21		88.22
	Wu & Liu [41]	TIP'21		97.48
	RTFM [38]	ICCV'21		97.21
	Ours: CLIP-TSA		CLIP	**98.32**

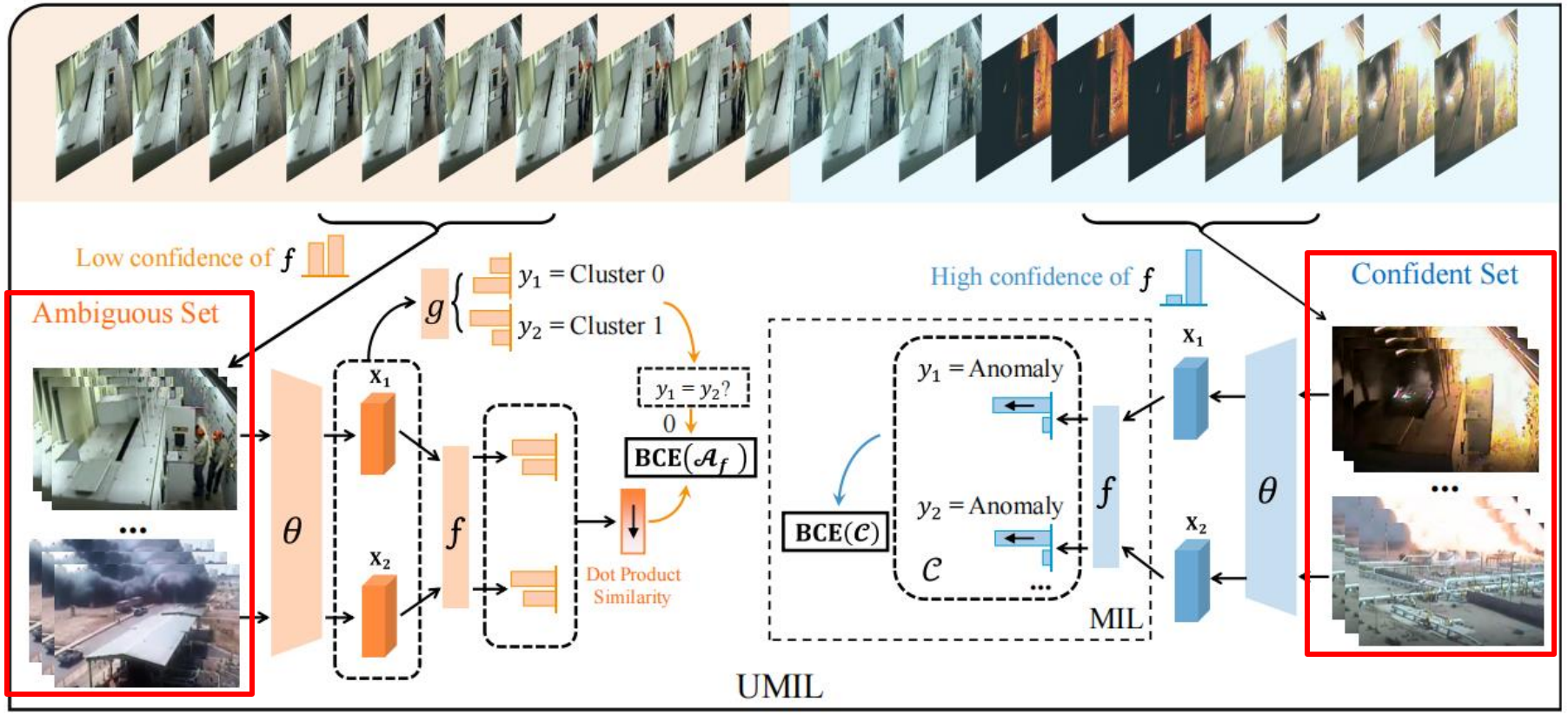
2 Overview of VLM-based VAD

Representation in the Early Exploration of VLMs

Method

In Unbiased MIL, the model is guided by a confidence set C to enhance prediction reliability and an ambiguous set A to reduce contextual bias through discriminative clustering.

ambiguous set A



confidence set C

Training with two confidence-based segment sets: ambiguous set A and confidence set C

Lv H, Yue Z, Sun Q, et al. Unbiased multiple instance learning for weakly supervised video anomaly detection. **CVPR**, 2023.

2 Overview of VLM-based VAD

Representation in the Early Exploration of VLMs

Result

On the UCF-Crime dataset, UMIL achieves the best performance among all methods, and on the TAD dataset, it significantly outperforms others in terms of AUC_A.

Category	Method	AUC_O (%)	AUC_A (%)
UVAD	SVM Baseline	50.00	50.00
	Conv-AE [6]	50.60	-
	Sohrab et al. [29]	58.50	-
	Lu et al. [18]	65.51	-
	BODS [33]	68.26	-
	GODS [33]	70.46	-
WSVAD	Sultani et al. [30]	75.41	54.25
	Zhang et al. [41]	78.66	-
	Motion-Aware [44]	79.10	62.18
	GCN-Anomaly [43]	82.12	59.02
	Wu et al. [34]	82.44	-
	RTFM [31]	84.30	-
	WSAL [22]	85.38	67.38
	Baseline	80.67	60.57
	UMIL	**86.75**	**68.68**

Category	Method	AUC_O (%)	AUC_A (%)
UVAD	SVM Baseline	50.00	50.00
	Luo *et al.* [19]	57.89	55.84
	Liu *et al.* [15]	69.13	55.38
WSVAD	Sultani *et al.* [30]	81.42	55.97
	Motion-Aware [44]	83.08	56.89
	GIG [21]	85.64	58.65
	WSAL [22]	89.64	61.66
	Baseline	89.10	56.47
	Ours	**92.93**	**65.82**

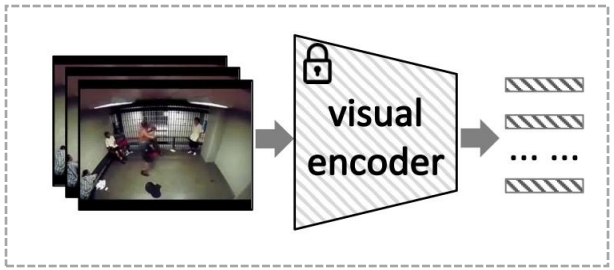
2 Overview of VLM-based VAD

Alignment in the Further Exploration of VLMs

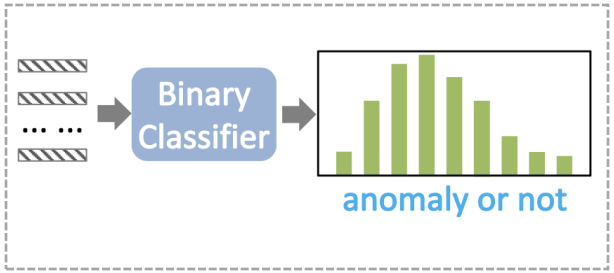
Problem

Current VAD methods mainly target closed-set scenarios and follow a "feature extraction + frame classification" paradigm, which presents several notable issues:

Issue 1 :
Lack of category interpretability



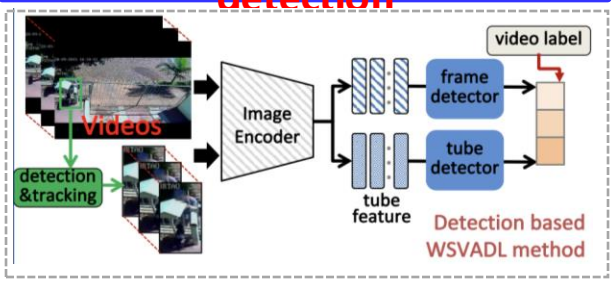
Feature Extraction



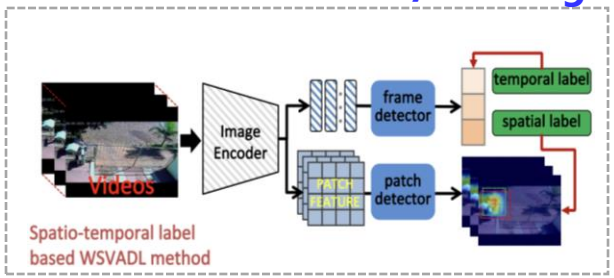
Frame Classification

Only outputs frame-level anomaly scores, lacking interpretability regarding the type or nature of the anomaly.

Issue 2 :
High computational complexity in spatiotemporal detection



Based on detection /tracking



Based on extra annotations

Relies on extra annotations or detection/tracking algorithms, with high computational cost.

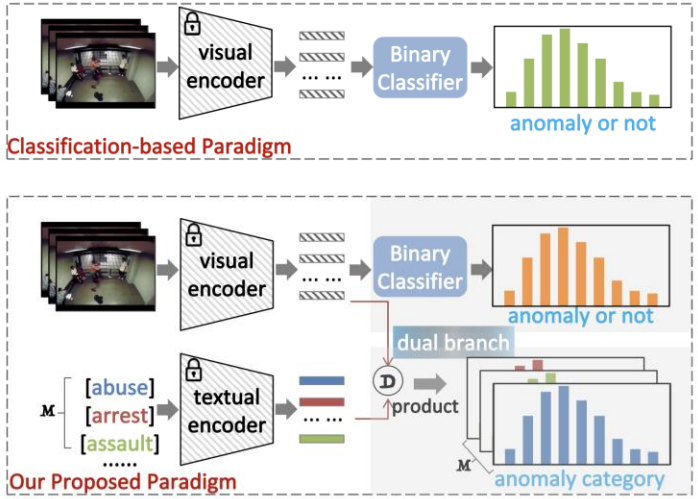
2 Overview of VLM-based VAD

Alignment in the Further Exploration of VLMs

Thought

Centered on cross-modal alignment, we propose a new VLM-based anomaly detection paradigm enhancing interpretability and spatiotemporal correspondence.

Addressing the lack of category interpretability



Transfer the pre-trained language–vision knowledge to the video anomaly detection task, thereby enhancing the interpretability of anomaly detection.

Addressing the high complexity of spatiotemporal detection



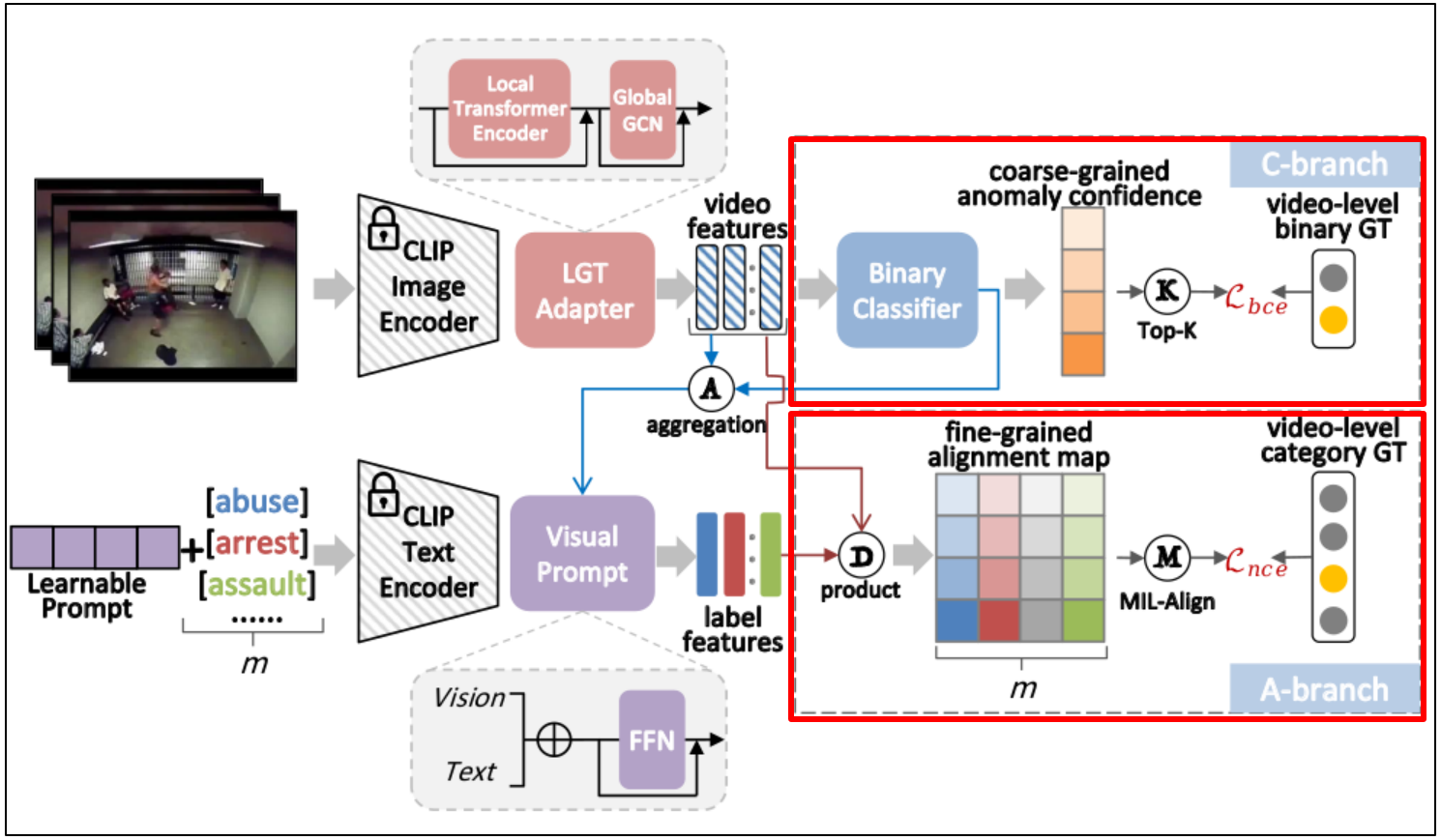
Leverage the VLM's inherent capability of mapping images to concepts to construct suitable spatio-temporal prompts, enabling efficient matching of anomalous behaviors across temporal and spatial dimensions.

2 Overview of VLM-based VAD

Alignment in the Further Exploration of VLMs

Method

VadCLIP employs a dual-branch network that detects video anomalies through visual classification and vision-language alignment. VadCLIP achieves interpretable anomaly detection by identifying anomaly categories.



Classification Branch

Alignment Branch

Wu P, et al. Vadclip: Adapting vision-language models for weakly supervised video anomaly detection. **AAAI**, 2024.

2 Overview of VLM-based VAD

Alignment in the Further Exploration of VLMs

Result

VadCLIP significantly outperforms previous approaches in fine-grained anomaly detection.

Method	mAP@IOU(%)						mAP@IOU(%)					
	0.1	0.2	0.3	0.4	0.5	AVG	0.1	0.2	0.3	0.4	0.5	AVG
Random	1.82	0.92	0.48	0.23	0.09	0.71	0.21	0.14	0.04	0.02	0.01	0.08
Sultani et al.	22.72	15.57	9.98	6.20	3.78	11.65	5.73	4.41	2.69	1.93	1.44	3.24
AVVD	30.51	25.75	20.18	14.83	9.79	20.21	10.27	7.01	6.25	3.42	**3.29**	6.05
VadCLIP	**37.03**	**30.84**	**23.38**	**17.90**	**14.31**	**24.70**	**11.72**	**7.83**	**6.40**	**4.53**	2.93	**6.68**

Method	AUC(%)	Ano-AUC(%)
SVM baseline	50.10	50.00
OCSVM (1999)	63.20	51.06
Hasan et al. (2016)	51.20	39.43
Ju et al. (2022)	84.72	62.60
Sultani et al. (2018)	84.14	63.29
Wu et al. (2020)	84.57	62.21
AVVD (2022)	82.45	60.27
RTFM (2021)	85.66	63.86
DMU (2023)	86.75	68.62
UMIL (2023)	86.75	68.68
CLIP-TSA (2023)	87.58	N/A
VadCLIP (Ours)	**88.02**	**70.23**

Category	Method	AP(%)
Semi	SVM baseline	50.80
	OCSVM (1999)	28.63
	Hasan et al. (2016)	31.25
Weak	Ju et al. (2022)	76.57
	Sultani et al. (2018)	75.18
	Wu et al. (2020)	80.00
	RTFM (2021)	78.27
	AVVD (2022)	78.10
	DMU (2023)	82.41
	CLIP-TSA (2023)	82.17
	VadCLIP (Ours)	**84.51**

Across different backbones, the proposed method not only achieves superior performance but also surpasses 88% AUC on the UCF-Crime dataset for the first time.

2 Overview of VLM-based VAD

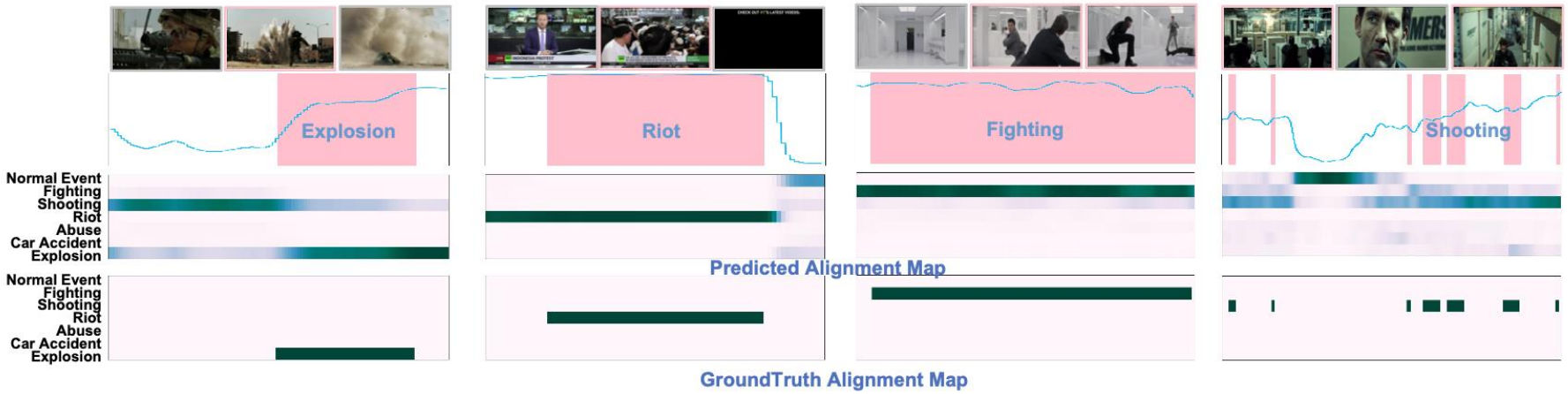
Alignment in the Further Exploration of VLMs

Result

VadCLIP significantly outperforms previous approaches in fine-grained anomaly detection.



Qualitative results of coarse-grained detection



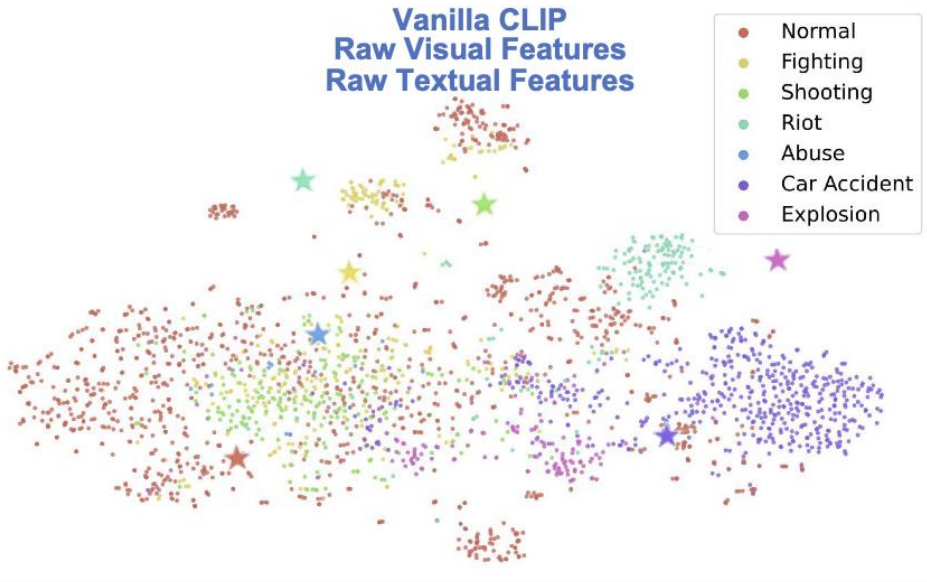
Qualitative results of fine-grained detection

2 Overview of VLM-based VAD

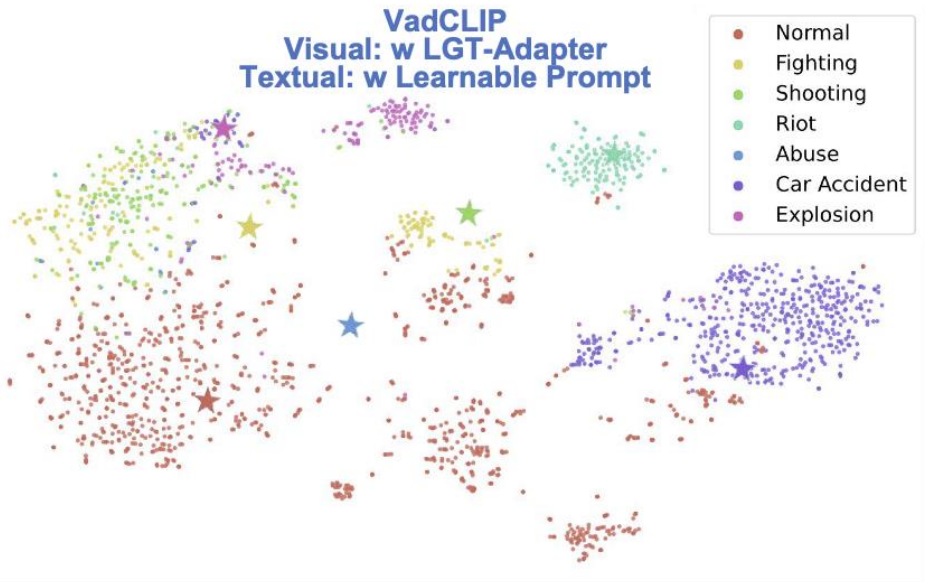
Alignment in the Further Exploration of VLMs

Result

VadCLIP significantly outperforms previous approaches in fine-grained anomaly detection.



Original CLIP feature visualization



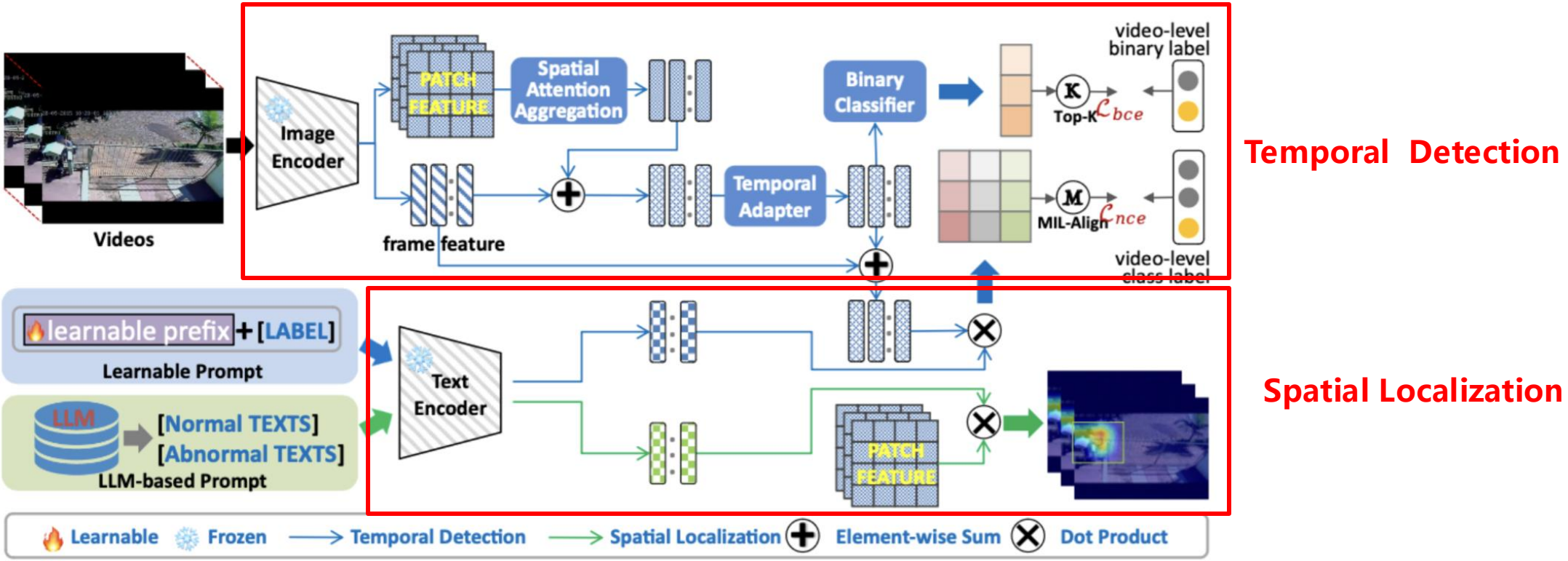
Optimized CLIP feature visualization

2 Overview of VLM-based VAD

Alignment in the Further Exploration of VLMs

Method

Decouple spatio-temporal anomaly detection into temporal detection and spatial localization. For temporal detection, irrelevant backgrounds are filtered out based on motion priors; for spatial localization, anomaly awareness is achieved in a training-free manner through a large language model.



Wu P, et al. Weakly Supervised Video Anomaly Detection and Localization with Spatio-Temporal Prompts. **ACM MM**, 2024.

2 Overview of VLM-based VAD

Alignment in the Further Exploration of VLMs

Result

It achieves state-of-the-art performance in both temporal and spatial anomaly detection, while demonstrating an order-of-magnitude advantage in model compactness and computational speed.

SOTA

Method	Feature	AUC(%)	TIoU(%)
Two Stream [50]	Two-stream	51.20	2.20
TSN [60]	TSN	53.20	2.60
C3D [56]	C3D	70.10	7.20
T-C3D [32]	C3D	74.50	10.20
ARTNet [58]	ARTNets	75.10	11.40
3DResNet [16]	I3D-ResNet	77.50	10.30
NLN [62]	I3D-ResNet	78.90	12.20
Liu et al. [33]	I3D-ResNet	82.00	16.40
SVM baseline	CLIP	50.10	N/A
OCSVM [49]	CLIP	63.20	N/A
Hasan et al. [17]	CLIP	51.20	N/A
Ju et al. [25]	CLIP	84.72	N/A
Sultani et al. [52]	CLIP	84.14	N/A
Sultani et al.† [52]	CLIP	67.11	16.82
Wu et al. [67]	CLIP	84.57	N/A
AVVD [68]	CLIP	82.45	N/A
RTFM [55]	CLIP	85.66	N/A
DMU [79]	CLIP	86.75	N/A
UMIL [39]	CLIP	86.75	N/A
CLIP-TSA [24]	CLIP	87.58	N/A
VadCLIP [71]	CLIP	88.02	22.05
STPrompt	CLIP	**88.08**	**23.90**

Method	Feature	AUC(%)	TIoU(%)
Georgescu et al. [12]	None	59.30	N/A
Georgescu et al. [12]+anomalies	None	61.30	N/A
Sultani et al. [52]	CLIP	53.23	N/A
Sultani et al.† [52]	CLIP	51.95	5.04
Wu et al. [67]	CLIP	53.70	N/A
RTFM [55]	CLIP	60.94	N/A
DMU [79]	CLIP	59.91	N/A
OPVAD [70]	CLIP	62.94	N/A
VadCLIP [71]	CLIP	62.32	3.67
STPrompt	CLIP	**63.98**	**8.17**

Method	Feature	Parameter	FLOPS
Zhong et al. [78]	C3D	78M	386.2G
RTFM [55]	I3D	28M	186.9G
SSRL [28]	I3D	191M	214.6G
SSRL* [28]	I3D	136M	214.6G
STPrompt	CLIP	**31.5M**	**44.8G**

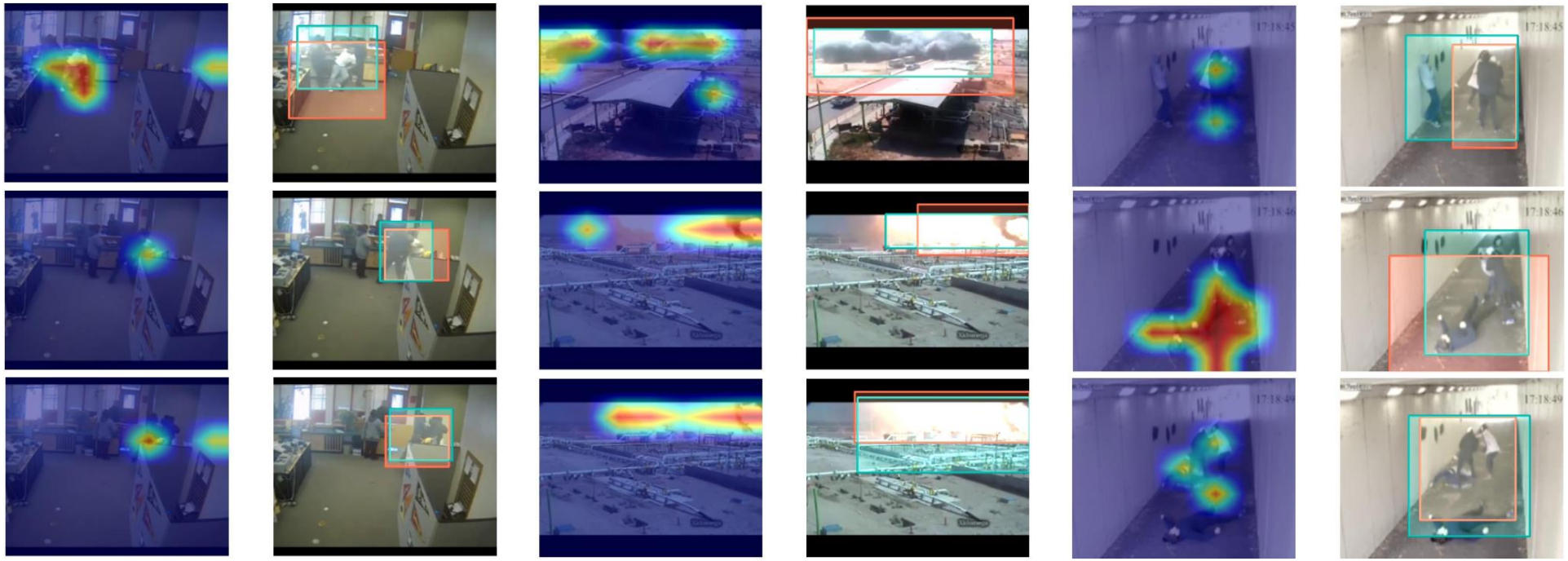
Lightweight and Fast

2 Overview of VLM-based VAD

Alignment in the Further Exploration of VLMs

Result

It achieves state-of-the-art performance in both temporal and spatial anomaly detection, while demonstrating an order-of-magnitude advantage in model compactness and computational speed.



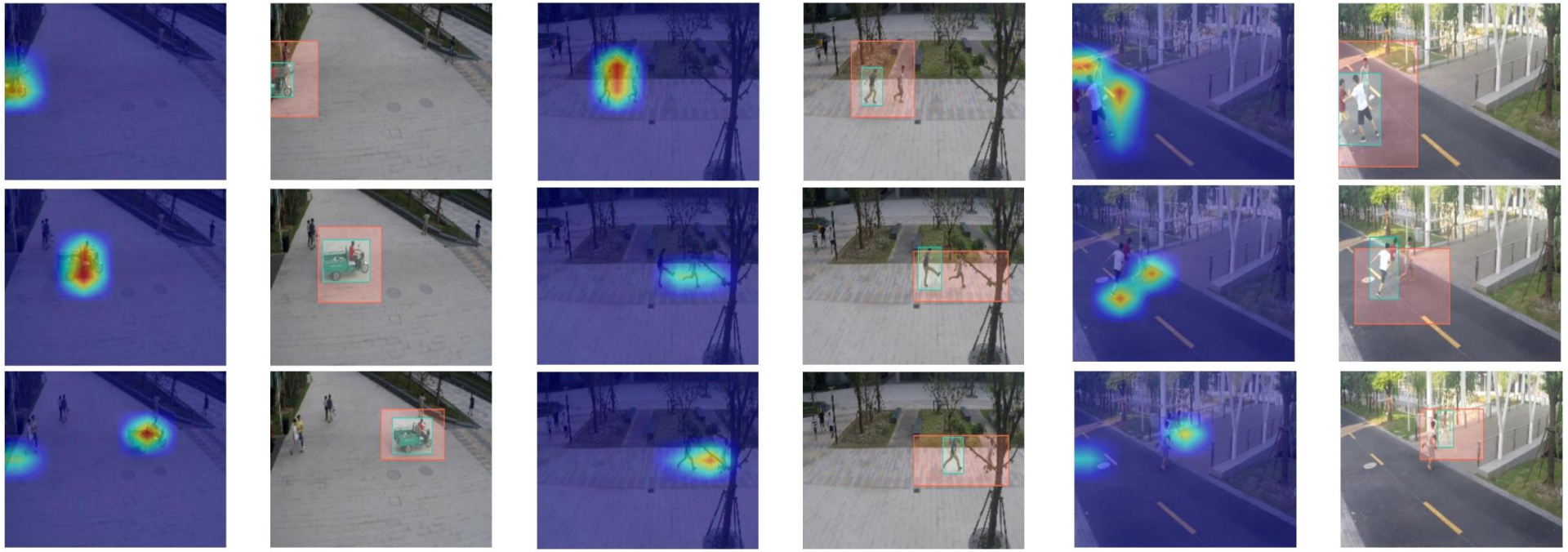
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2 Overview of VLM-based VAD

Alignment in the Further Exploration of VLMs

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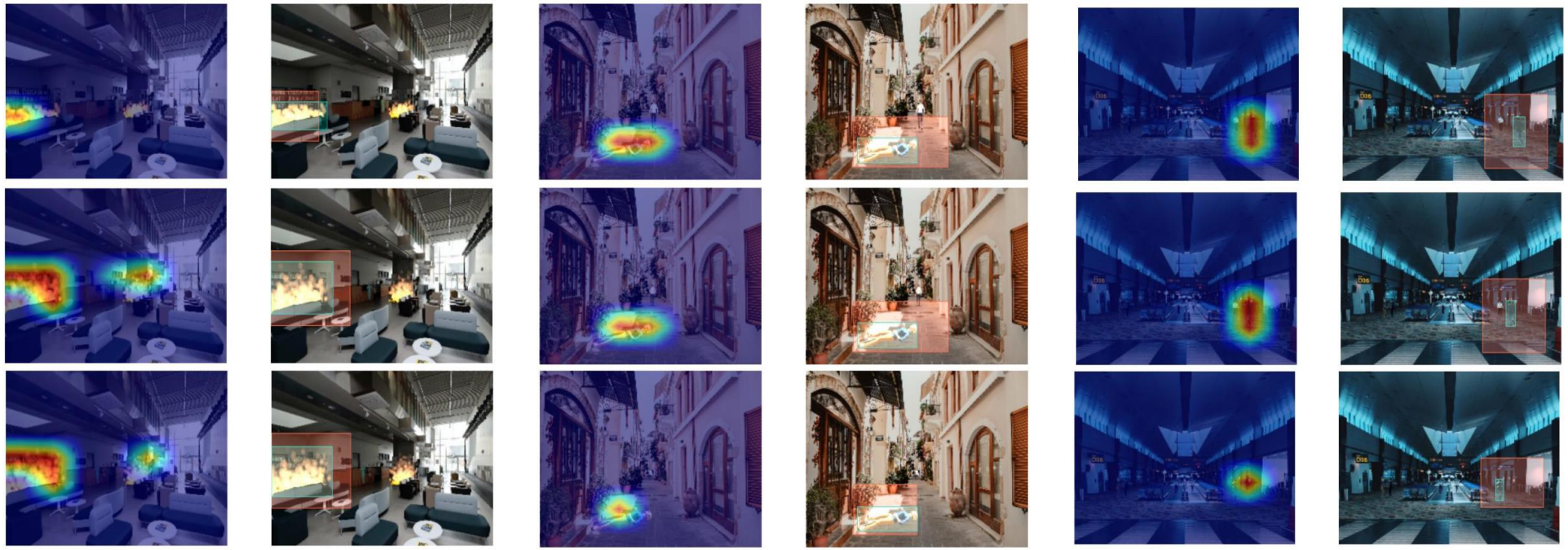


2 Overview of VLM-based VAD

Alignment in the Further Exploration of VLMs

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It achieves state-of-the-art performance in both temporal and spatial anomaly detection, while demonstrating an order-of-magnitude advantage in model compactness and computational speed.



scene4_1 scene17_7 scene19_5

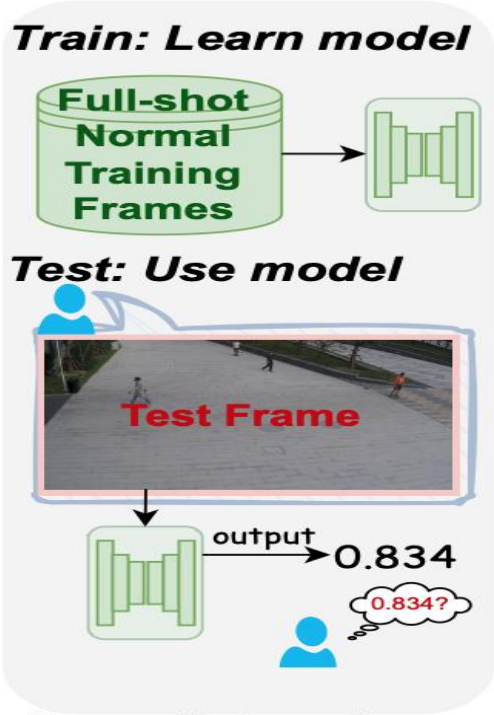
2 Overview of VLM-based VAD

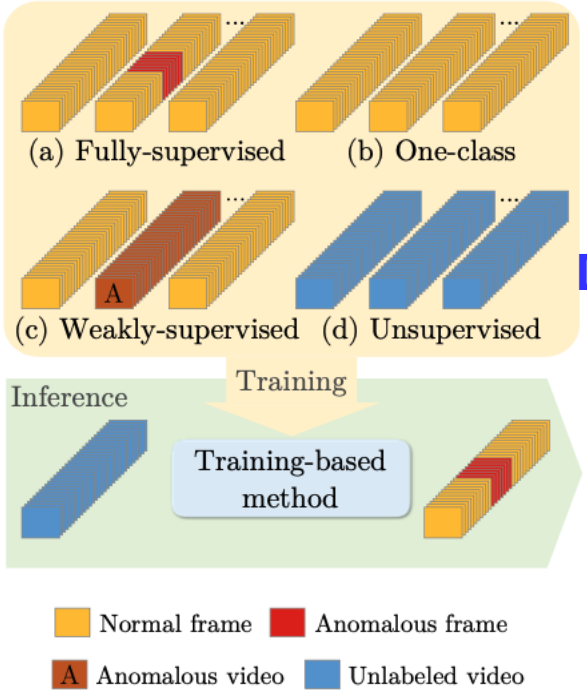
Reasoning in the Advanced Exploration of VLMs

Proble m

Traditional VAD methods mostly rely on closed-set labels and classifiers, making it difficult to provide semantic explanations for abnormal behaviors. Moreover, many of these methods depend on supervised training, resulting in limited generalization when faced with insufficient data or open-world scenarios.。

Lack of semantic interpretability





Dependence on extensive annotated data

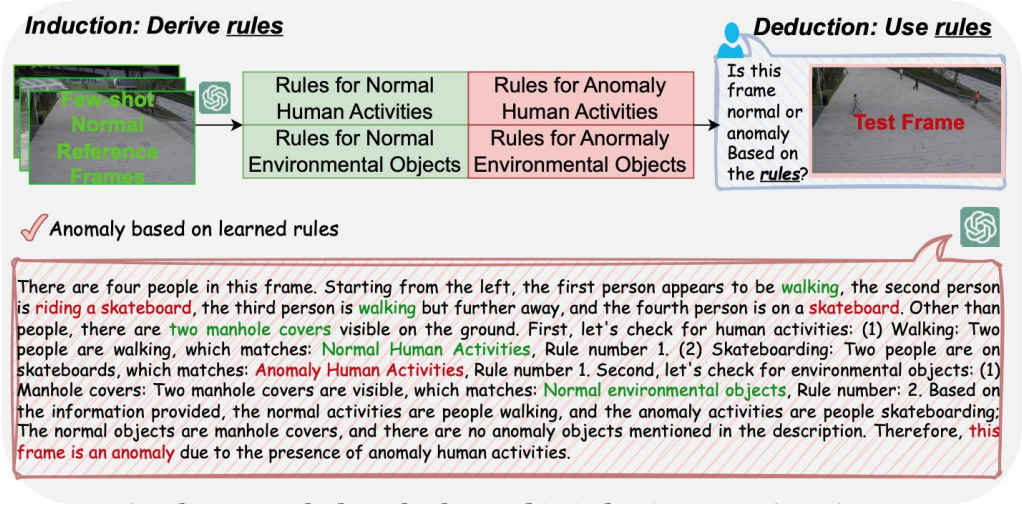
2 Overview of VLM-based VAD

Reasoning in the Advanced Exploration of VLMs

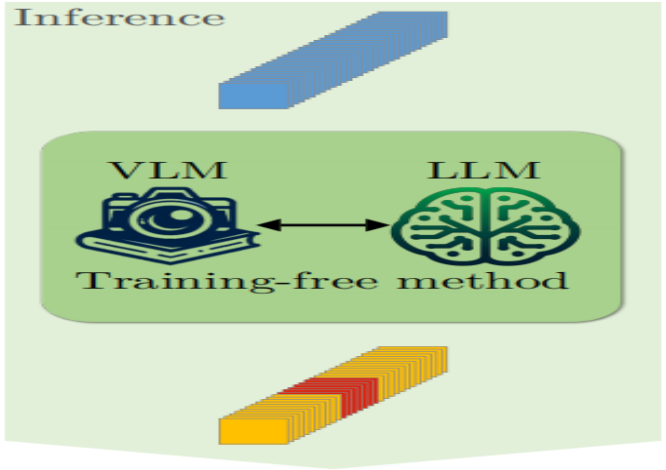
Thought

Fully leverage the reasoning ability and general knowledge of pre-trained VLMs and LLMs to provide interpretations of anomalous behaviors, enabling a training-free anomaly detection framework that reduces dependence on annotated data.

Inductive reasoning based on VLM and LLM



Leveraging general knowledge of VLMs and LLMs.

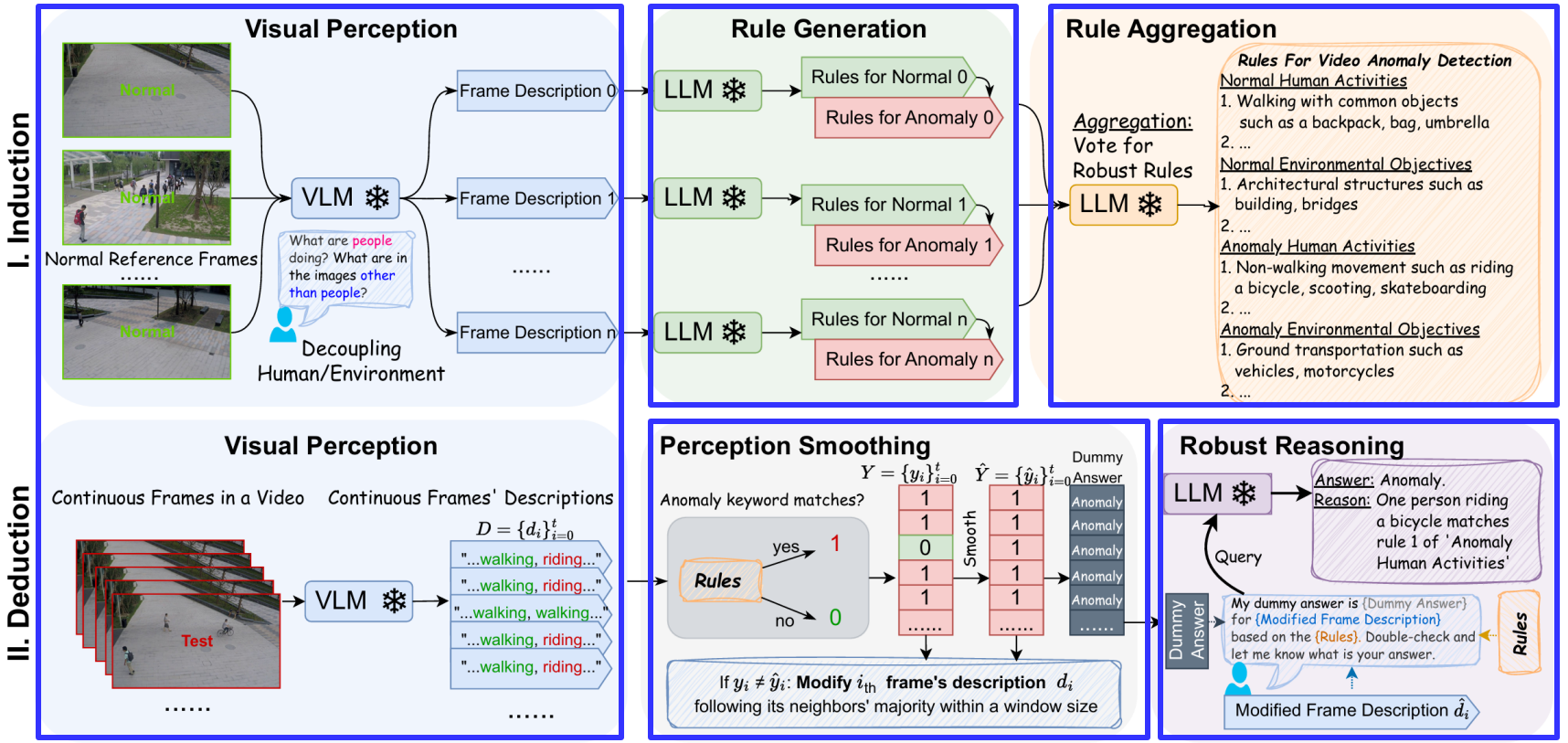


2 Overview of VLM-based VAD

Reasoning in the Advanced Exploration of VLMs

Method

AnomalyRuler framework consists of inductive and deductive stages. Through multi-module collaboration, it detects video anomalies, leveraging the two-stage design to enable rule-based inference for



Induction

+

Deduction

Yang Y, et al. Follow the rules: reasoning for video anomaly detection with large language models. **ECCV**, 2024.

2 Overview of VLM-based VAD

Reasoning in the Advanced Exploration of VLMs

Result

AnomalyRuler achieves state-of-the-art performance across four VAD benchmark datasets, reaching 85.2% AUC on the ShT dataset.

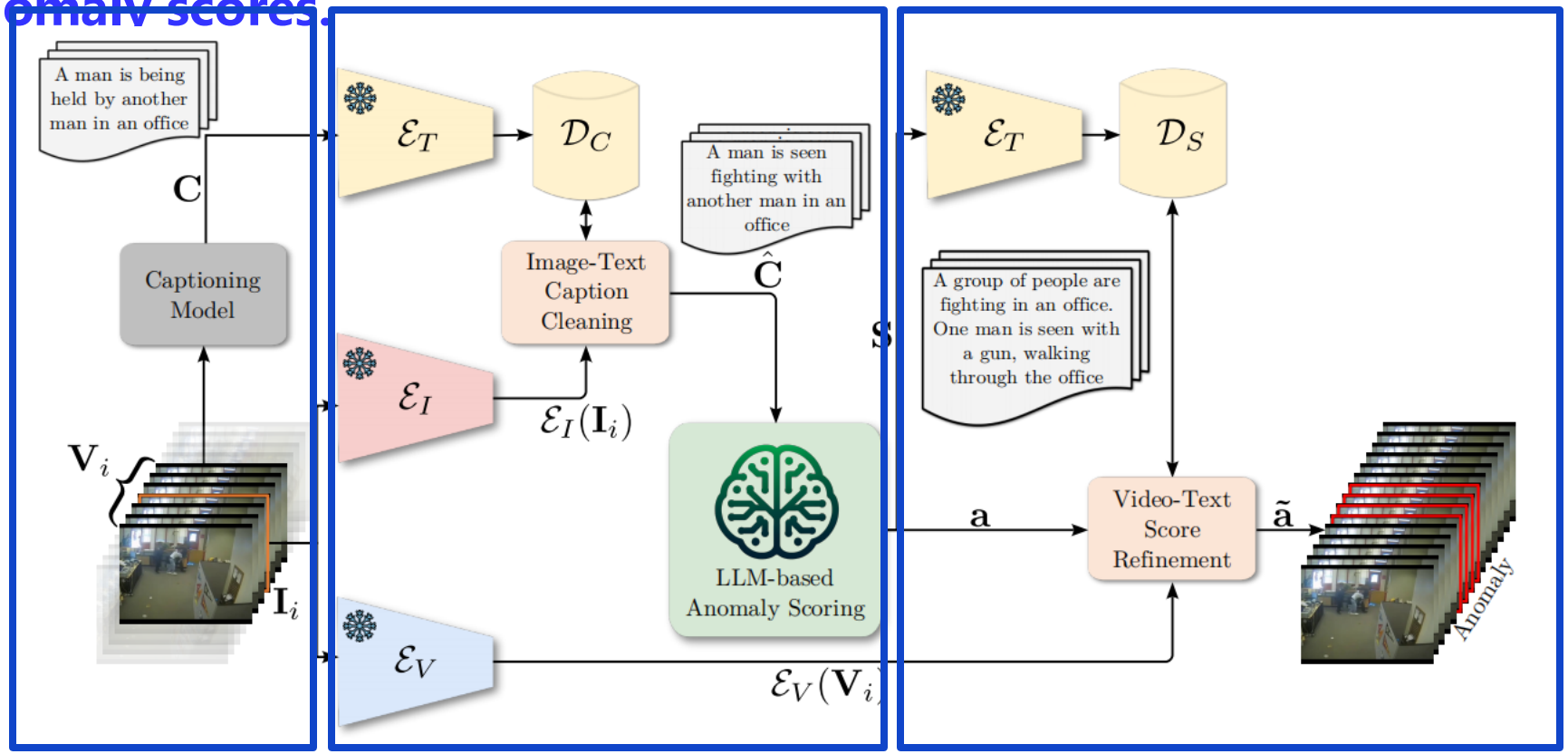
Method	Venue	Image Only	Training	Ped2	Ave	ShT	UB
MNAD [36]	CVPR-20	✓	✓	97.0	88.5	70.5	-
rGAN [29]	ECCV-20	✓	✓	96.2	85.8	77.9	-
CDAE [9]	ECCV-20	✓	✓	96.5	86.0	73.3	-
MPN [30]	CVPR-21	✓	✓	96.9	89.5	73.8	-
NGOF [50]	CVPR-21	✗	✓	94.2	88.4	75.3	-
HF2 [25]	ICCV-21	✗	✓	**99.2**	91.1	76.2	-
BAF [14]	TPAMI-21	✗	✓	98.7	92.3	82.7	59.3
GCL [56]	CVPR-22	✗	✓	-	-	79.6	-
S3R [53]	ECCV-22	✗	✓	-	-	80.5	-
SSL [49]	ECCV-22	✗	✓	99.0	**92.2**	84.3	-
zxVAD [3]	WACV-23	✗	✓	96.9	-	71.6	-
HSC [45]	CVPR-23	✗	✓	98.1	93.7	83.4	-
FPDM [54]	ICCV-23	✓	✓	-	90.1	78.6	62.7
SLM [43]	ICCV-23	✓	✓	97.6	90.9	78.8	-
STG-NF [18]	ICCV-23	✗	✓	-	-	**85.9**	71.8
AnomalyRuler-base	-	✓	✗	96.5	82.2	84.6	69.8
AnomalyRuler	-	✓	✗	97.9	89.7	85.2	**71.9**

2 Overview of VLM-based VAD

Reasoning in the Advanced Exploration of VLMs

Method

LAVAD generates initial captions through a captioning module, which are then processed by key components including I2T caption cleaning, LLM-based anomaly scoring, and V2T score refinement, ultimately producing frame-level anomaly scores.



Harnessing Large Language Models for Training-free Video Anomaly Detection. **CVPR**, 2024.

2 Overview of VLM-based VAD

Reasoning in the Advanced Exploration of VLMs

Result

LAVAD outperforms both one-class and unsupervised methods, achieving a 6.08% improvement on UCF-Crime and a 17.03% gain over RareAnom on XD-Violence

METHOD	BACKBONE	AUC(%)
SULTANI *et al.* [24]	C3D-RGB	75.41
SULTANI *et al.* [24]	I3D-RGB	77.92
IBL [41]	C3D-RGB	78.66
GCL [40]	ResNext	79.84
GCN [42]	TSN-RGB	82.12
MIST [5]	I3D-RGB	82.30
WU *et al.*[36]	I3D-RGB	82.44
CLAWS [39]	C3D-RGB	83.03
RTFM [28]	VideoSwin-RGB	83.31
RTFM [28]	I3D-RGB	84.03
WU & LIU [35]	I3D-RGB	84.89
MSL [15]	I3D-RGB	85.30
MSL [15]	VideoSwin-RGB	85.62
S3R [34]	I3D-RGB	85.99
MGFN [2]	VideoSwin-RGB	86.67
MGFN [2]	I3D-RGB	86.98
SSRL [13]	I3D-RGB	87.43
CLIP-TSA [11]	ViT	87.58
SVM [24]	-	50.00
SSV [23]	-	58.50
BODS [33]	I3D-RGB	68.26
GODS [33]	I3D-RGB	70.46
GCL [40]	ResNext	74.20
TUR *et al.* [30]	ResNet	65.22
TUR *et al.* [31]	ResNet	66.85
DYANNET [27]	I3D	79.76
ZS CLIP [22]	ViT	53.16
ZS IMAGEBIND (IMAGE) [6]	ViT	53.65
ZS IMAGEBIND (VIDEO) [6]	ViT	55.78
LLAVA-1.5 [17]	ViT	72.84
LAVAD	ViT	**80.28**

UCF-Crime

METHOD	BACKBONE	AP(%)	AUC(%)
WU *et al.* [36]	C3D-RGB	67.19	-
WU *et al.* [36]	I3D-RGB	73.20	-
MSL [15]	C3D-RGB	75.53	-
WU AND LIU[35]	I3D-RGB	75.90	-
RTFM [28]	I3D-RGB	77.81	-
MSL [15]	I3D-RGB	78.28	-
MSL [15]	VideoSwin-RGB	78.58	-
S3R[34]	I3D-RGB	80.26	-
MGFN [2]	I3D-RGB	79.19	-
MGFN [2]	VideoSwin-RGB	80.11	-
HASAN *et al.* [8]	AE^{RGB}	-	50.32*
LU *et al.* [19]	Dictionary	-	53.56*
BODS [33]	I3D-RGB	-	57.32*
GODS[33]	I3D-RGB	-	61.56*
RAREANOM [26]	I3D-RGB	-	68.33*
ZS CLIP [22]	ViT	17.83	38.21
ZS IMAGEBIND (IMAGE) [6]	ViT	27.25	58.81
ZS IMAGEBIND (VIDEO) [6]	ViT	25.36	55.06
LLAVA-1.5 [17]	ViT	50.26	79.62
LAVAD	ViT	**62.01**	**85.36**

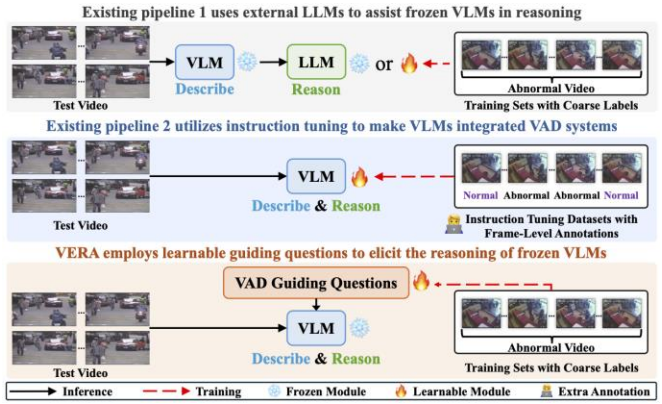
XD-Violence

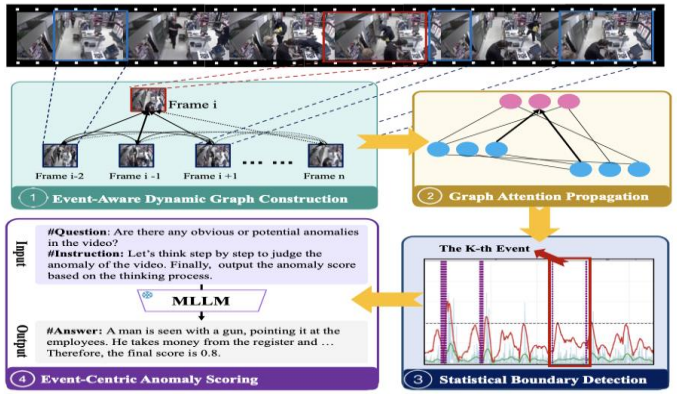
2 Overview of VLM-based VAD

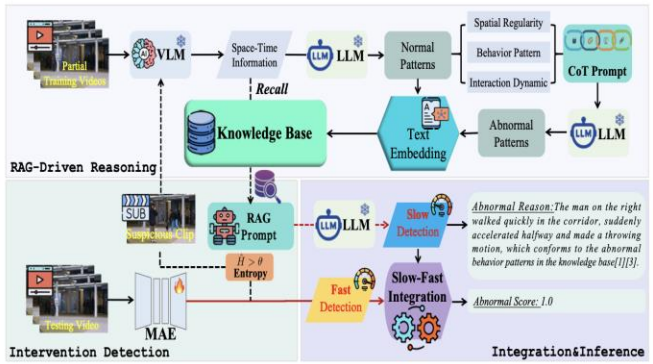
Reasoning in the Advanced Exploration of VLMs

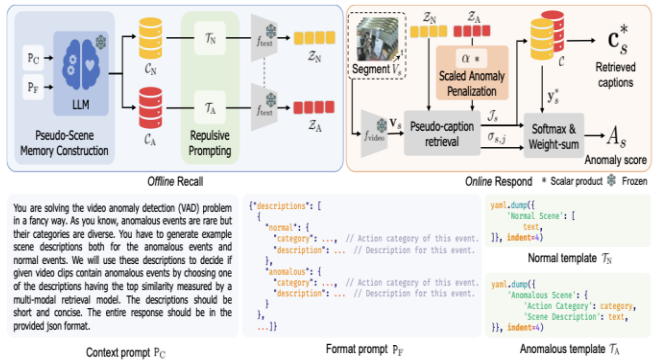
Method

During the same period, there are also many VLM-based reasoning approaches for anomaly detection, such as VERA, EventVAD, SlowFastVAD, and Flashback.







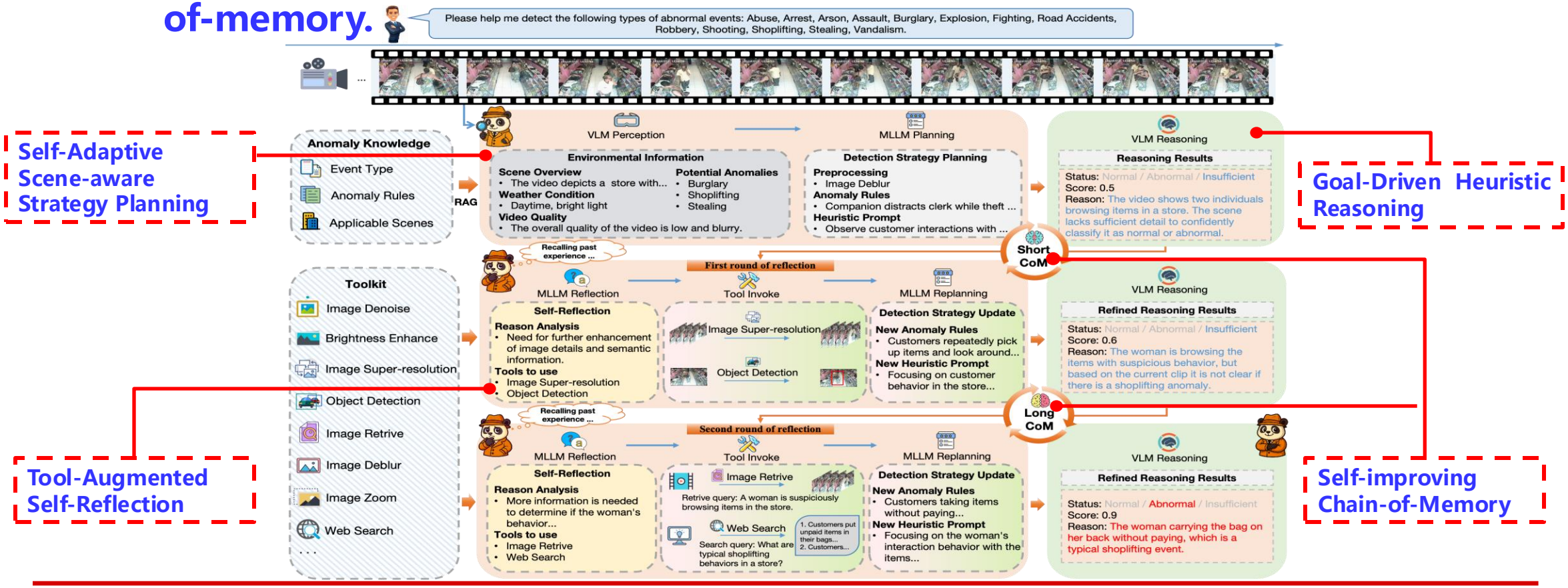


2 Overview of VLM-based VAD

Reasoning in the Advanced Exploration of VLMs

Method

PANDA, as an agentic AI engineer for generalized VAD, it is equipped with four key capabilities: 1. Self-adaptive scene-aware strategy planning; 2. Goal-driven heuristic reasoning; 3. Tool-augmented self-reflection; 4. Self-improving chain-of-memory.



Yang Z et al. PANDA: Towards Generalist Video Anomaly Detection via Agentic AI Engineer. NeurIPS 2025.

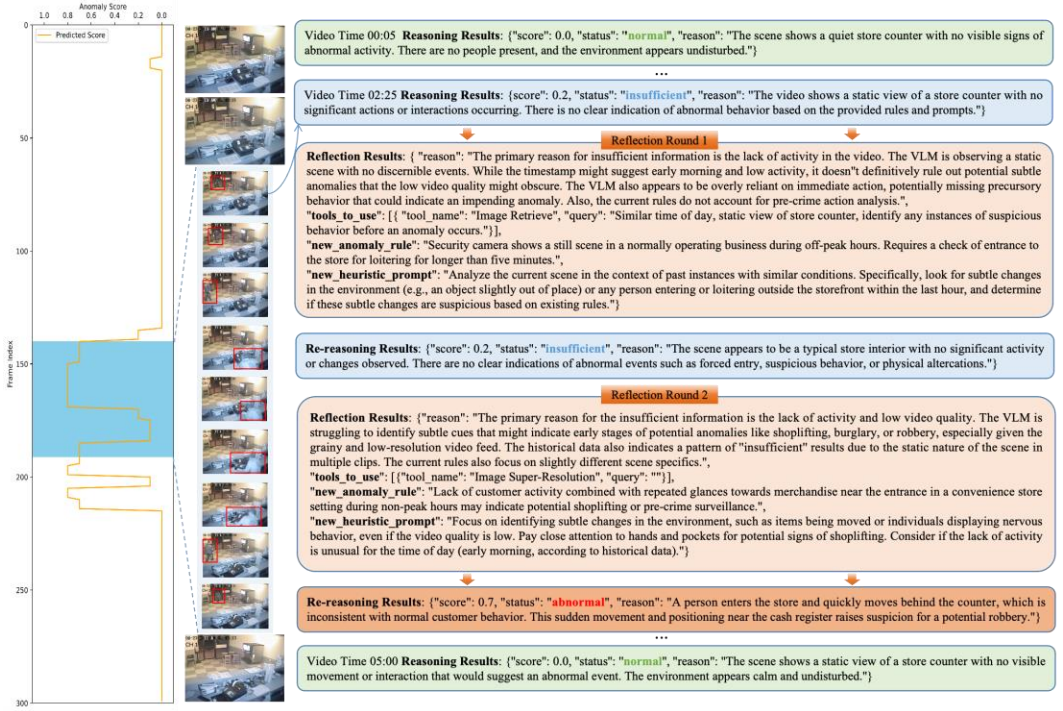
2 Overview of VLM-based VAD

Reasoning in the Advanced Exploration of VLMs

Result

PANDA significantly outperforms all existing training-free baselines across all four datasets, even under online settings.

Methods		Supervision	Expl.	Manual-free	Mode	Multi-Scenario UCF (AUC%)	Multi-Scenario XD (AP%)	Open-Set UB (AUC%)	Complex Scenario CSAD (AUC%)
Specialized methods									
AED-MAE [23]	[CVPR2024]	Semi	✗	✗	Offline	-	-	58.50	-
STPAG [35]	[CVPR2024]	Semi	✗	✗	Offline	-	-	57.98	-
HL-Net [16]	[ECCV2020]	Weak	✗	✗	Offline	82.44	73.67	-	-
RTFM [36]	[ICCV2021]	Weak	✗	✗	Offline	84.30	77.81	64.94	-
UR-DMU [37]	[AAAI2023]	Weak	✗	✗	Offline	86.97	81.66	59.91	-
VadCLIP [15]	[AAAI2024]	Weak	✗	✗	Offline	88.02	84.51	-	-
TPWNG [14]	[CVPR2024]	Weak	✗	✗	Offline	87.79	83.68	-	-
VERA [38]	[CVPR2025]	Weak	✓	✗	Offline	86.55	70.11	-	64.52
Holmes-VAU [22]	[CVPR2025]	Instru-Tuned	✓	✗	Offline	88.96	87.68	56.77	72.47
ZS CLIP [26]	[ICML2021]	Training-free	✓	✗	Offline	53.16	17.83	46.2	32.45
LLAVA-1.5 [26]	[CVPR2024]	Training-free	✓	✗	Offline	72.84	50.26	53.71	47.78
LAVAD [29]	[CVPR2024]	Training-free	✓	✗	Offline	80.28	62.01	64.23	57.26
AnomalyRuler [30]	[ECCV2024]	Training-free	✓	✗	Offline	-	-	71.90	-
Generalized method									
PANDA	**(ours)**	Training-free	✓	✓	Offline	**84.89**	**70.16**	**75.78**	**73.12**
					Online	82.57	63.57	72.41	71.25

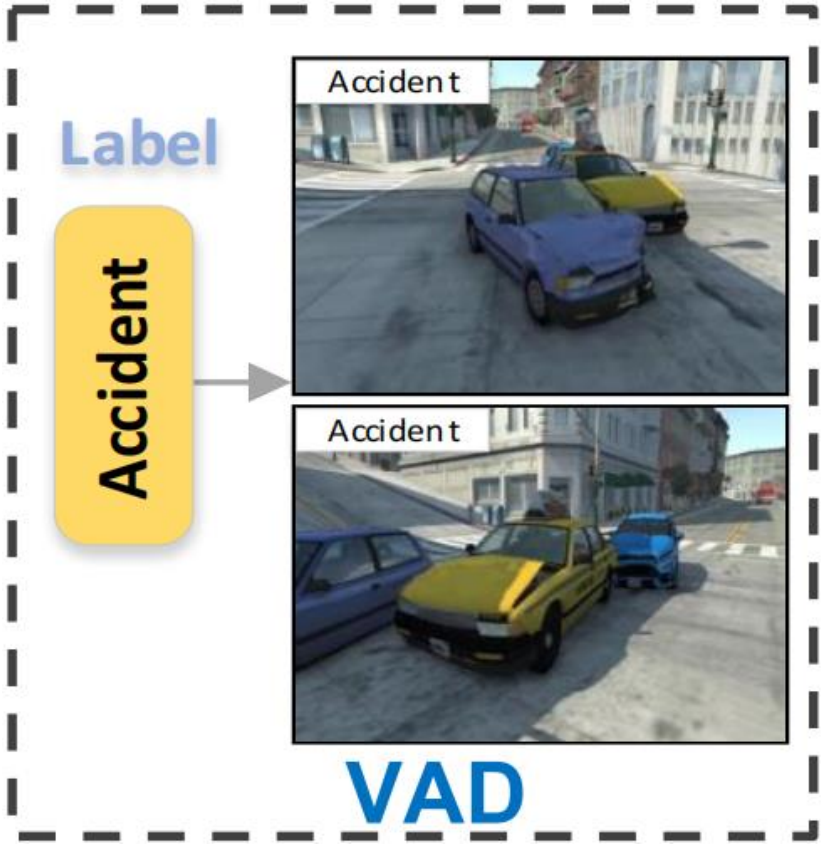


Yang Z et al. PANDA: Towards Generalist Video Anomaly Detection via Agentic AI Engineer. NeurIPS 2025.

3 Overview of VLM-based VAR

Problem

VAD only provides fuzzy single labels, making it difficult to accurately express complex event sequences. In reality, users prefer to retrieve corresponding abnormal videos through detailed text/audio descriptions.



VAD provides single label



Traditional video retrieval tasks

The single label provided by traditional VAD cannot describe in detail the abnormal events that occur in the video.

The event duration of traditional video retrieval tasks is generally short.

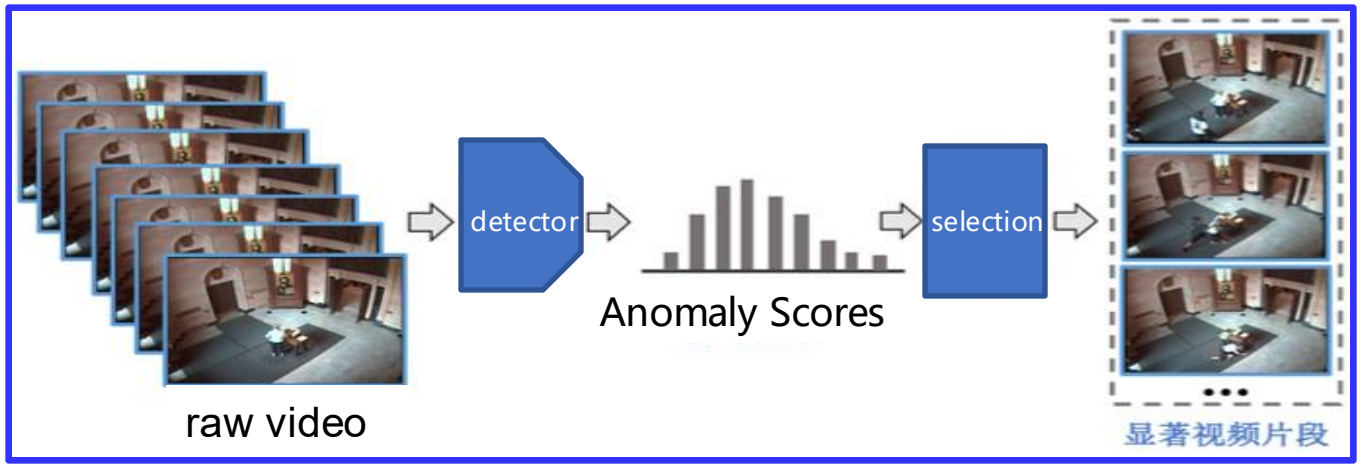
3 Overview of VLM-based VAR

Thought

Propose VAR task that can retrieve relevant anomaly event segments from untrimmed long videos through text or audio cross modal queries, enhancing practicality and expressiveness.

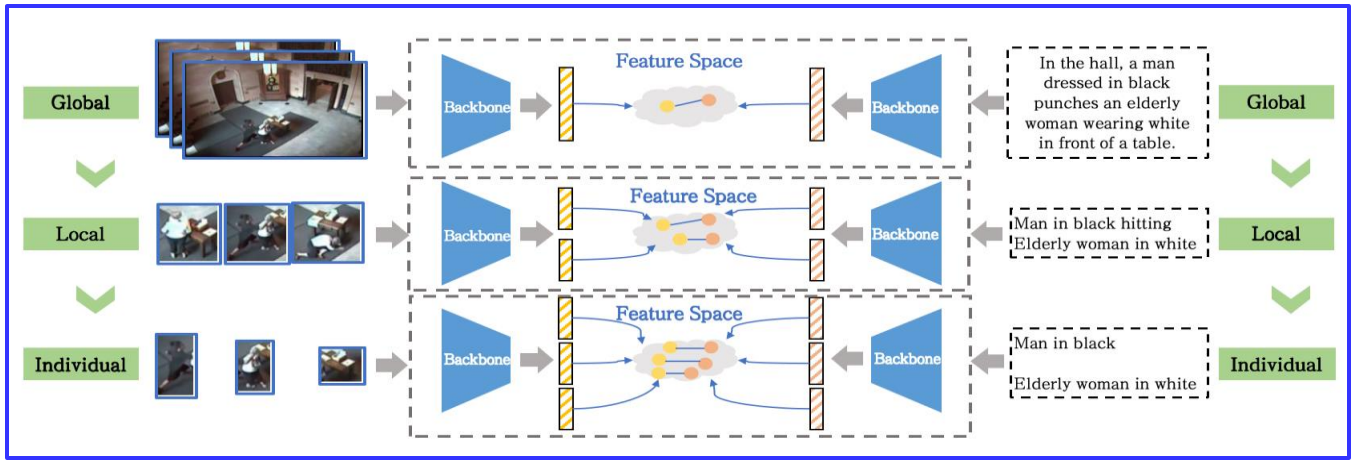
Method1: Abnormal prior assisted retrieval

Anomaly scores guide the model to focus on abnormal segments, enhancing key perception and retrieval performance. Core idea: use VAD to aid retrieval.



Method2: Multi-granularity information alignment

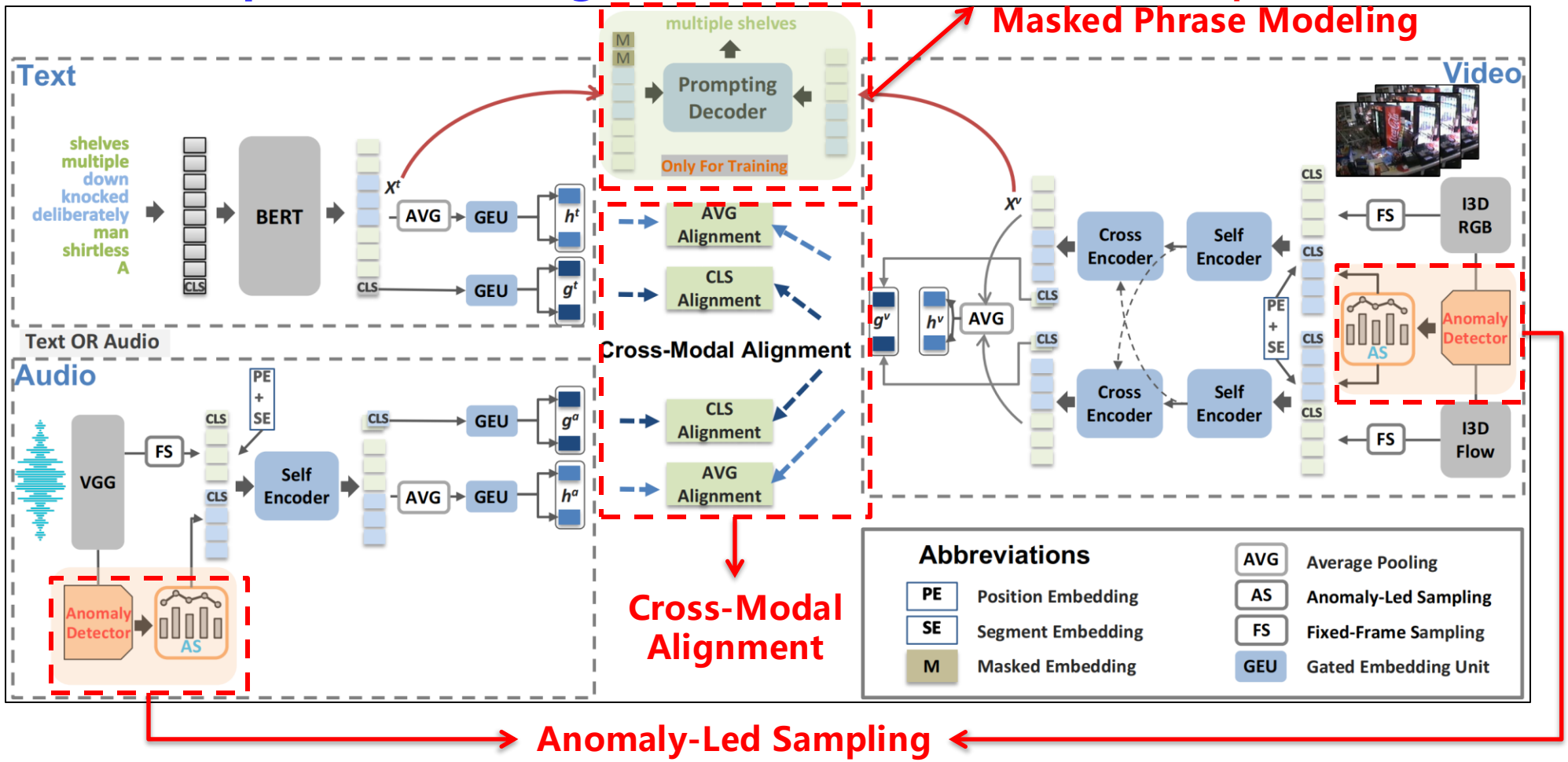
Align video and text/audio at global, local, and individual levels to optimize information com-pleteness and computational efficiency.



3 Overview of VLM-based VAR

Method

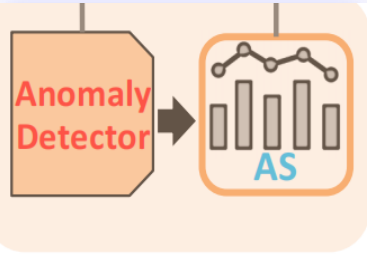
Introduce anomaly-led sampling to focus on key abnormal segments, and enhance fine-grained video-text alignment through video prompt based masked phrase modeling.



3 Overview of VLM-based VAR

Method

Introduce anomaly-led sampling to focus on key abnormal segments, and enhance fine-grained video-text alignment through video-prompt-based masked phrase modeling.



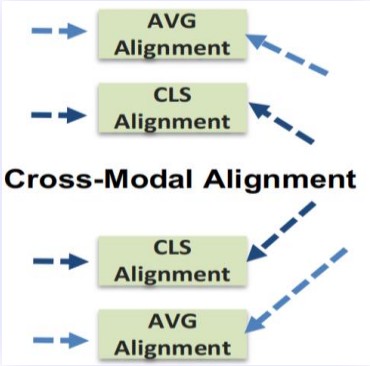
Module1: Anomaly-Led Sampling

Use confidence scores from the anomaly detector to select key frame segments, enhancing the model's focus on abnormal regions.



Module2: Video Prompt Based Masked Phrase Modeling

Use the video as a prompt to guide BERT in predicting masked phrases, enhancing fine-grained video-text alignment.



Module3: Cross-Modal Alignment

Dual-level alignment mechanism: CLS-level (classification token) and AVG-level (average fusion) alignment.

Wu P, et al. Toward video anomaly retrieval from video anomaly detection: New benchmarks and model. **TIP**, 2024.

3 Overview of VLM-based VAR

Result

As the first method for VAR, it achieves significant improvements over previous approaches on two newly constructed large-scale datasets, UCFCrime-AR and XDViolence-AR.

TABLE II
COMPARISONS WITH THE STATE-OF-THE-ART METHODS ON UCFCRIME-AR.

Method	Text→Video				Video→Text				SumR↑
	R@1↑	R@5↑	R@10↑	MdR↓	R@1↑	R@5↑	R@10↑	MdR↓	
Random Baseline	0.3	2.1	3.4	144.0	0.3	1.0	3.1	145.5	10.2
CE [12]	6.6	19.7	32.4	23.5	5.5	19.7	32.4	21.0	116.3
MMT [13]	8.3	26.2	39.3	16.0	7.2	23.1	39.0	16.0	143.1
T2VLAD [89]	7.6	23.4	39.7	15.5	6.2	**27.9**	43.1	14.0	147.9
X-CLIP [58]	8.2	27.2	41.7	16.0	6.9	25.8	40.3	15.0	150.1
HL-Net [7]	5.5	20.2	38.3	19.5	5.5	22.8	35.5	20.0	127.8
XML [74]	6.9	24.1	42.4	14.0	6.6	25.9	43.4	13.0	149.3
ALAN	**9.0**	**27.9**	**44.8**	**14.0**	**7.3**	24.8	**46.9**	**12.0**	**160.7**

TABLE III
COMPARISONS WITH THE STATE-OF-THE-ART METHODS ON XDVIOLENCE-AR.

Method	Audio→Video				Video→Audio				SumR↑
	R@1↑	R@5↑	R@10↑	MdR↓	R@1↑	R@5↑	R@10↑	MdR↓	
Random Baseline	0.4	0.6	2.5	399.5	0.1	0.6	0.8	399.5	5.0
CE [12]	11.4	33.3	47.0	12.5	13.0	34.3	46.4	13.0	185.4
MMT [13]	20.5	53.5	68.0	5.0	23.0	54.6	69.5	5.0	289.1
T2VLAD [89]	22.4	56.1	71.0	4.0	23.2	57.1	73.5	4.0	303.3
X-CLIP [58]	26.4	61.1	73.9	3.0	26.4	61.3	73.8	4.0	322.9
HL-Net [7]	12.4	36.6	48.3	11.0	13.4	38.3	52.1	10.0	201.1
XML [74]	22.9	55.6	70.3	5.0	22.6	57.4	71.4	4.0	300.2
ALAN	**29.8**	**68.0**	**82.0**	**3.0**	**32.3**	**70.0**	**82.3**	**3.0**	**364.4**

TABLE IV
COMPARISONS OF DIFFERENT SAMPLINGS ON UCFCRIME-AR.

Sampling	Text→Video		Video→Text	
	R@1↑	R@10↑	R@1↑	R@10↑
FS (N=50)	6.6	35.5	4.8	42.4
AS (N=50)	7.9	37.6	5.5	41.7
FS (N=100)	6.6	37.6	6.2	40.3
FS+AS (N=50)	**9.0**	**44.8**	**7.3**	**46.9**

TABLE V
COMPARISONS OF DIFFERENT SAMPLINGS ON XDVIOLENCE-AR.

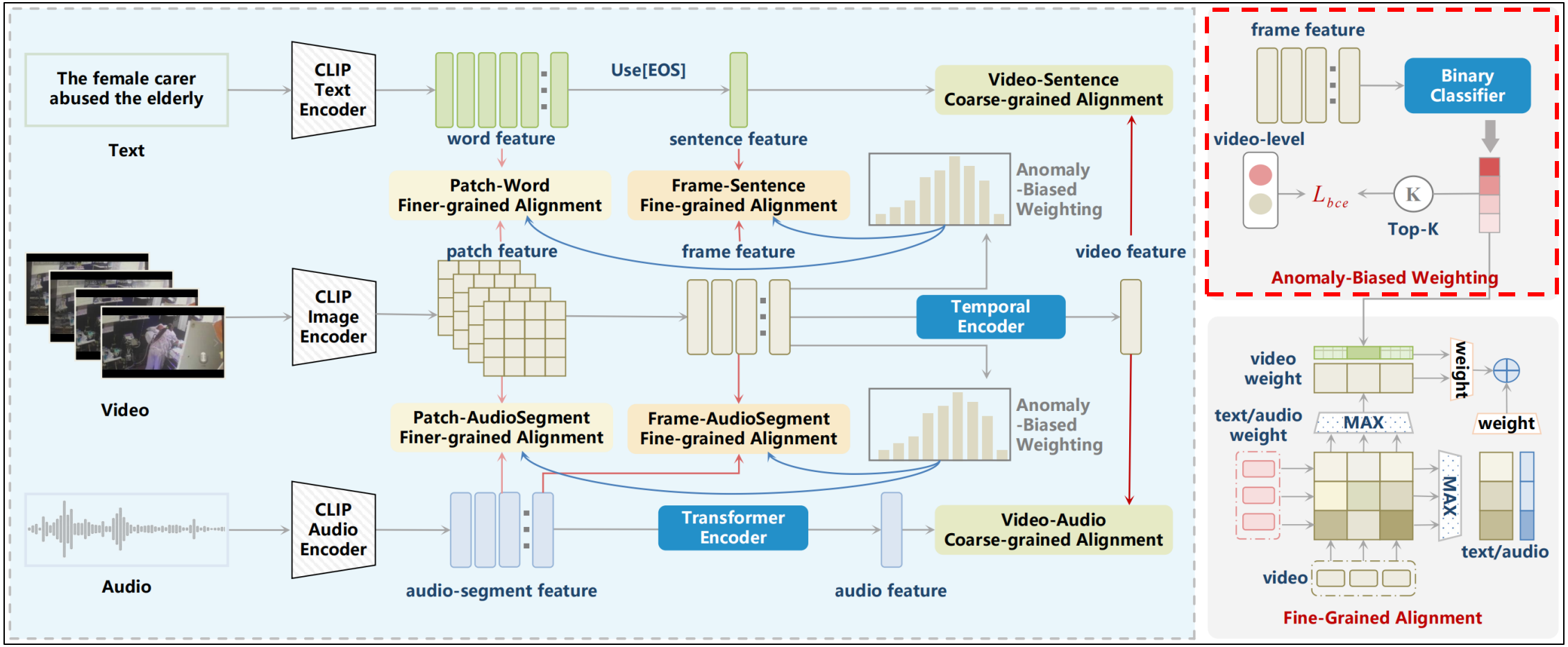
Sampling	Text→Video		Video→Text	
	R@1↑	R@10↑	R@1↑	R@10↑
FS (N=50)	29.6	80.4	31.1	80.9
AS (N=50)	26.9	78.6	27.4	78.9
FS (N=100)	28.5	81.0	29.8	81.8
FS+AS (N=50)	**29.8**	**82.0**	**32.3**	**82.3**

Wu P, et al. Toward video anomaly retrieval from video anomaly detection: New benchmarks and model. **TIP**, 2024.

3 Overview of VLM-based VAR

Method

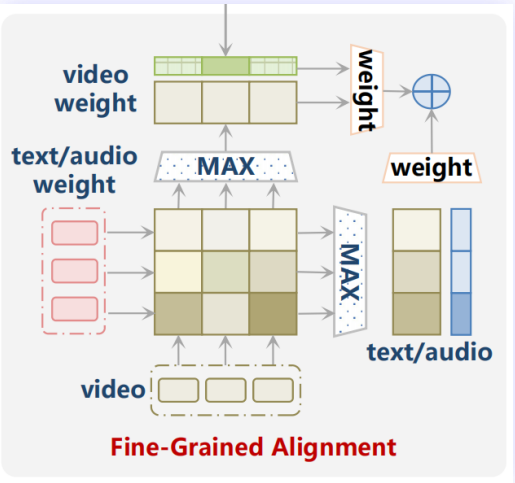
Achieve global, local, and individual alignment from both video-text and video-audio perspectives, while introducing an anomaly-biased weighting mechanism to focus on key abnormal segments.



3 Overview of VLM-based VAR

Method

Achieve global, local, and individual alignment from both video-text and video-audio perspectives, while introducing an anomaly-biased weighting mechanism to focus on key abnormal segments.

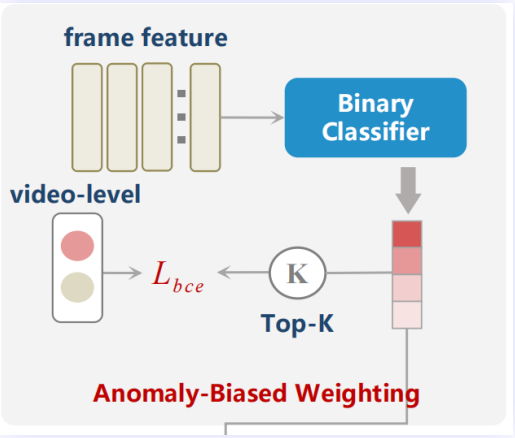


Module1: Unified Hierarchical Alignment

Three-level alignment strategy:

- Global: video-sentence/audio for overall semantics.
- Loca: frame-sentence/audio segment for temporal context.
- Individual: patch-word/audio segment for fine details.

Achieves cross-modal semantic alignment across multiple granularities.



Module2: Anomaly-Biased Weighting Mechanism

- A lightweight detector generates frame-level anomaly confidence.
- Dynamic weighting is applied based on anomaly probability.
- Emphasize abnormal segments and suppress normal ones.

Focus on key frames to improve alignment accuracy.

Wu P, et al. VarCMP: Adapting Cross-Modal Pre-Training Models for Video Anomaly Retrieval. **AAAI**, 2025.

3 Overview of VLM-based VAR

Result

The proposed method achieves state-of-the-art performance on two large-scale datasets, demonstrating the effectiveness of multi-granularity information alignment.

Method	Text → Video				Video → Text			
	R@1↑	R@5↑	R@10↑	MdR↓	R@1↑	R@5↑	R@10↑	MdR↓
Non-CLIP Methods								
Random Baseline	0.3	2.1	3.4	144.0	0.3	1.0	3.1	145.5
CE(2019)	6.6	19.7	32.4	23.5	5.5	19.7	32.4	21.0
MMT(2020)	8.3	26.2	39.3	16.0	7.2	23.1	39.0	16.0
HL-Net(2020)	5.5	20.2	38.3	19.5	5.5	22.8	35.5	20.0
XML(2020)	6.9	24.1	42.4	14.0	6.6	25.9	43.4	13.0
T2VLAD(2021)	7.6	23.4	39.7	15.5	6.2	27.9	43.1	14.0
X-CLIP*(2022)	8.2	27.2	41.7	16.0	6.9	25.8	40.3	15.0
ALAN(2024a)	9.0	27.9	44.8	14.0	7.3	24.8	46.9	12.0
CLIP-based Methods								
CLIP4Clip(2022)	17.8	40.1	57.5	8.0	15.3	33.0	44.6	13.0
TS2-Net(2022)	23.4	54.1	70.7	4.0	17.7	53.7	68.0	5.0
UCoFiA(2023)	23.6	53.8	69.2	5.0	22.8	54.1	67.7	4.0
ALAN†(2024a)	10.3	36.9	57.9	8.0	14.1	42.4	57.9	8.0
VarCMP(Ours)	**28.6**	**60.3**	**74.8**	**4.0**	**24.1**	**56.8**	**72.1**	**4.0**

Table 1: Comparisons with the state-of-the-art methods on UCFCrime-AR.

Coarse	Fine	Finer	Text → Video		Video → Text	
			R@1↑	R@10↑	R@1↑	R@10↑
√			24.7	72.3	20.7	64.6
√	√		26.0	72.9	21.8	68.0
√	√	√	**28.6**	**74.8**	**24.1**	**72.1**

Table 3: The effect of different level alignments on UCFCrime-AR.

Coarse	Fine	Finer	Audio → Video		Video → Audio	
			R@1↑	R@10↑	R@1↑	R@10↑
√			31.4	86.1	28.6	87.0
√	√		34.0	90.2	33.1	89.2
√	√	√	**38.2**	**92.5**	**38.5**	**91.1**

Table 4: The effect of different level alignments on XDViolence-AR.

Wu P, et al. VarCMP: Adapting Cross-Modal Pre-Training Models for Video Anomaly Retrieval. **AAAI**, 2025.

4 Overview of VLM-based VAC

A Preliminary Exploration of VLM Alignment and Generation

Proble m

Existing methods can only provide coarse-grained descriptions and fail to finely distinguish and characterize the specific behavioral features of each individual, making it difficult to perform further situational analysis.



Video-level: A man in dark clothes enters a convenience store, threatens the clerk with a weapon, and flees after the robbery.

Event-level:

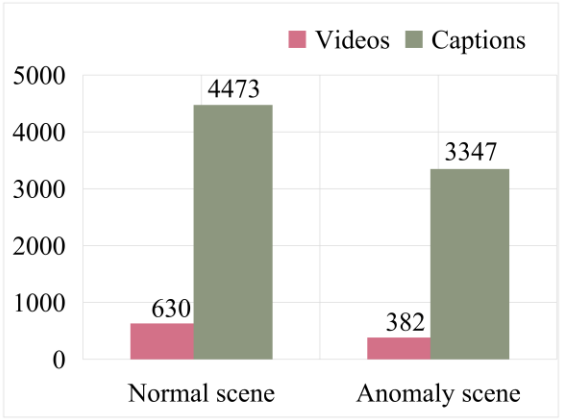
00.0–05.3: A man in a gray shirt and white helmet looks around the counter.
10.3–15.6: A black-haired man in a blue-striped shirt enters.
15.6–18.2: The gray-shirted man follows, draws a knife, and threatens him.
19.8–20.7: He signals a robbery.
20.7–23.1: The blue-shirted man grabs a knife from under the counter.
24.2–29.9: The gray-shirted man flees; the blue-shirted man chases him off-screen.

4 Overview of VLM-based VAC

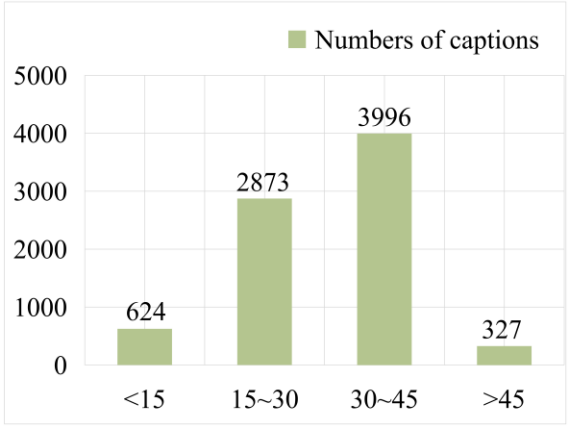
A Preliminary Exploration of VLM Alignment and Generation

Thought

Construct an individual-centric dataset with 1,012 videos and 7,820 annotations to provide fine-grained behavior descriptions.



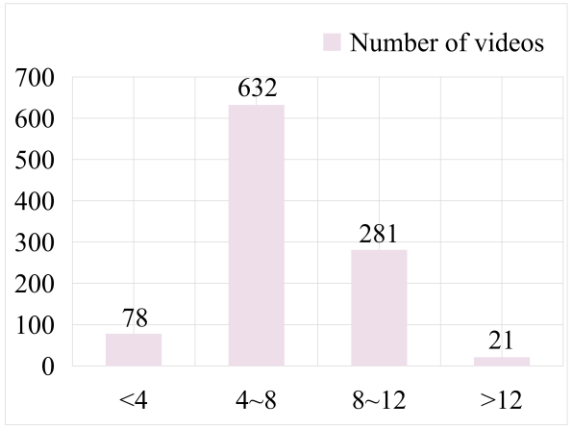
Distribution of normal and abnormal scenes



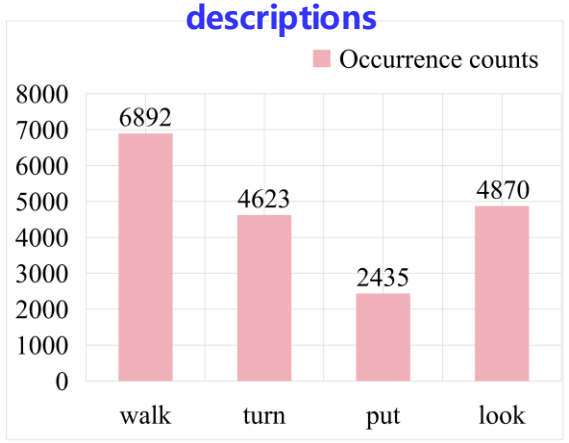
Word count statistics of pedestrian descriptions



[0, 12.52]Outdoor, a lady wearing a black jacket and black jeans was walking on the pavement. She was hit from the back by 2 and fell face first on the pavement. She laid there unconscious. [0.66, 7.74]Outdoor, a man in a black jacket and black pants was walking after 1 on the pavement. He was holding something in his right hand. He ran to 1 and hit 1's head with his right hand. He then turned around and jogged away.



Number of pedestrians per video



Number of pedestrian actions

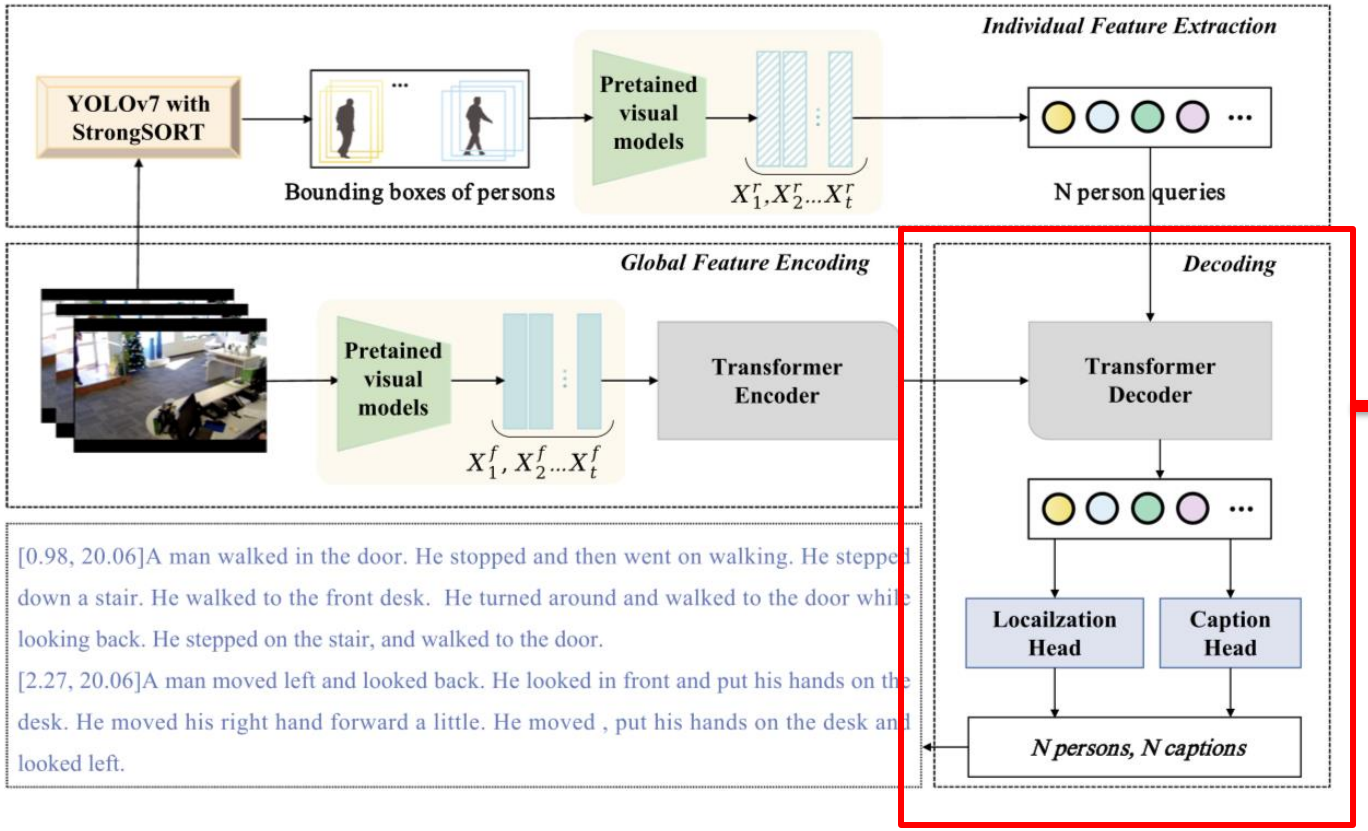
Sample statistics display

4 Overview of VLM-based VAC

A Preliminary Exploration of VLM Alignment and Generation

Method

By integrating individual and global features, the decoder extracts representations to locate the time segments of each individual and generate corresponding behavior descriptions.



By decoding global and local features, a localization head identifies the individual' s time segments, while a description head generates behavior descriptions.

Zhou L, et al. Human-centric Behavior Description in Videos: New Benchmark and Model . **TMM**, 2024.

4 Overview of VLM-based VAC

A Preliminary Exploration of VLM Alignment and Generation

Result

This work achieves the best performance in individual behavior description and in both normal and abnormal scenes.

Method	Features	BLEU-4	CIDER	METEOR	ROUGE-L	Extra Data Training
Vid2Seq [74]	CLIP ViT-L/14	40.5	71.4	26.7	57.9	✓
VALOR [20]	CLIP/VideoSwin Transformer	34.6	62.3	19.4	48.7	✓
mPlug-2 [75]	Dual-vision Encoder	31.7	61.4	16.2	43.6	✗
VAST [76]	Vision Transformer	34.1	64.2	-	-	✓
GIT2 [77]	Florence	29.6	57.8	15.2	41.1	✓
VLAB [78]	CLIP	29.4	56.7	14.5	41.3	✓
MT [79]	TSN	32.8	60.2	16.3	44.5	✗
BMT [80]	I3D	34.3	62.7	18.6	46.8	✗
PDVC [33]	TSP	36.5	65.4	20.7	49.7	✗
Ours	C3D	43.6	73.8	28.5	57.1	✗
Ours	CLIP	44.2	74.6	29.2	57.9	✗
Ours	**I3D**	**44.7**	**75.2**	**30.3**	**59.2**	✗

Compared to the current state-of-the-art PDVC, it improves by 8.2%, 9.8%, 9.6%, and 9.5% across four metrics.

Method	Normal Scenarious				Anomaly Scenarious			
	BLEU-4	CIDER	METEOR	ROUGE-L	BLEU-4	CIDER	METEOR	ROUGE-L
VALOR [20]	34.6	62.3	19.4	48.7	33.6	59.3	16.2	47.7
VAST [76]	34.1	64.2	-	-	32.1	60.3	-	-
OURS	44.7	75.2	30.3	59.2	44.2	74.8	30.6	58.7

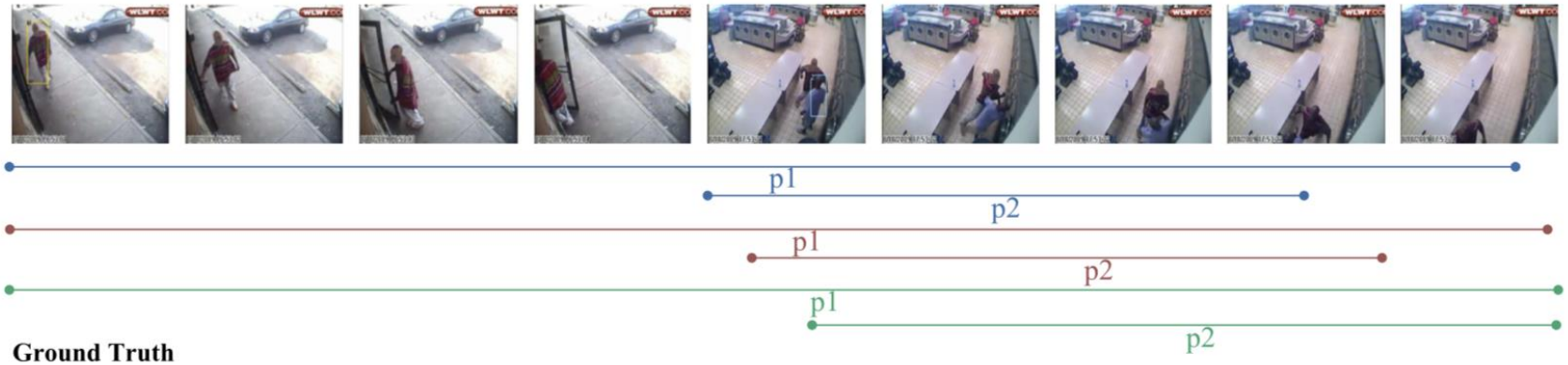
Compared with large models, our method also attains superior performance in both normal and abnormal scene descriptions.

4 Overview of VLM-based VAC

A Preliminary Exploration of VLM Alignment and Generation

Result

Identify each individual in the first frame, track them until they disappear, and objectively describe their entire sequence of actions.



Ground Truth

p1:[0, 7.92]Outdoor, a man in a red t-shirt with white stripes was walking on the pavement with his left hand inside his pants pocket. He pulled open a gate with his right hand and walked inside. Inside a laundromat, he swung his right hand and punched 2. He swung his left hand and punched 2 again. As 2 stumbled backward, he backed 2 towards the row of washing machines and kept punching 2. He then disappeared.

p2:[4.35, 6.98]Inside the laundromat, a man in a blue T-shirt was standing in front of a row of washing machines. He was being punched in the face by 1. He stumbled backward from the impact of the punch and hit his back on a washing machine. He had his arms stretched out in front of him towards 1 as he was being hit and backed away. He then disappeared.

Ours

p1:[0, 8.24]A man was walking on the road with his left hand down. He opened a door with his right hand and walked inside. He took his right hand and beat. He beat again. He moved forward and kept beating. He then was no longer visible.

p2:[4.68, 7.58]A man was standing in front of a machines. He was being beaten in the face by others. He moved backward and hit on a machine. He had his arms towards others when he was being beaten and backed away. He then was no longer visible.

Vid2Seq

p1:[0, 9.75]A man in a red T-shirt was walking on the road. He opened the door and entered. He beat a man again and again. He then went out.

p2:[5.25, 9.75]A man in a blue T-shirt was standing in front of cabinet. He was being beaten by others. He moved backward and hit on a machine. He beat the man and then couldn't visible.

4 Overview of VLM-based VAC

Prelude to VLM: Data Generation

Proble m

Current surveillance video tasks mainly focus on classifying and localizing predefined anomalies, providing only a coarse-grained understanding of abnormal video semantics and suffering from the following major issues:

Problem1:
Lacks event element parsing



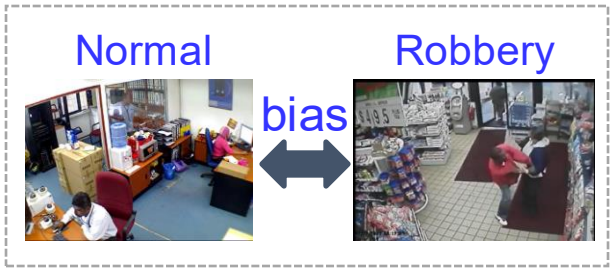
Missing event elements



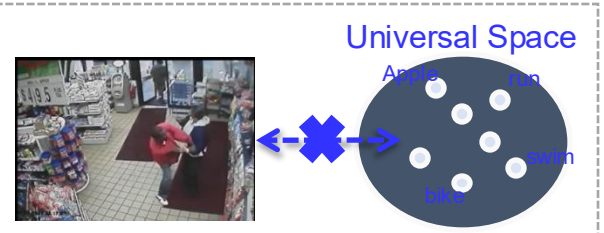
Limited interpretability

Focuses only on binary anomaly detection, failing to explain how anomalies occur.

Problem2:
Fails with general vision-language models



Biased visual features



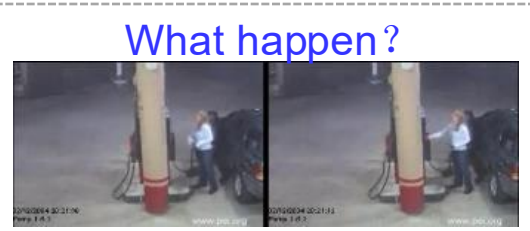
Weak semantic correlations

Significant domain gap exists between surveillance behavior patterns and general action semantics.

Problem3 :
Lacks fine-grained multimodal annotations.



Lack of temporal annotations



Missing semantic labels

Precise event timestamps and detailed natural language descriptions are lacking.

4 Overview of VLM-based VAC

Prelude to VLM: Data Generation

Thought We establish the first multimodal benchmark for surveillance video understanding, focusing on event parsing, domain adaptation, and cross-modal supervision.

Thought1: Event parsing



Woman falls; Man attacking; Man wearing black hood ⋯

Professional annotators provide detailed natural language descriptions of people, actions, scenes, and event sequences.

Thought2: Domain adaptation



Fully manual annotation ensures semantic accuracy and alignment with surveillance scenarios.

Thought3 : Cross-modal fine-grained supervision



[20.5, 30,1]: A car hit by a motorcycle

Create a triple mapping of video clips, textual descriptions, and temporal intervals to support detailed multimodal understanding of anomalies.

4 Overview of VLM-based VAC

Prelude to VLM: Data Generation

Method

UCA combines multimodal annotations with real surveillance scenarios to provide the first benchmark for semantic understanding and complex tasks, advancing from predefined event detection to open semantic understanding.

sentence query: A black car drove past the corner, and a gray car followed closely behind it.



start point |← ---------- →| end point

00:01.0 00:06.1

sentence query: The man in red took out his gun and shot at the striped man. At the same time, the four people sitting on the bench in the front stood up and hid in the corner of the room.



start point |← ---------- →| end point

01:34.4 01:45.6

Supports video summarization

Supports natural language video retrieval

Supports dense captioning

Supports multimodal anomaly detection

Multimodal alignment: precise temporal annotations with fine-grained natural language descriptions.

Yuan, et al. Towards surveillance video-and-language understanding: New dataset baselines and challenges. **CVPR**. 2024.

4 Overview of VLM-based VAC

Prelude to VLM: Data Generation

Result

The proposed UCA multimodal annotations improve traditional task performance and capture the ambiguity and semantic complexity of surveillance videos.

method	visual	General Swinbert	Surveillance Swinbert	AUC
Visual	✓	✗	✗	83.1%
TEVAD [8] (2023)	✓	✓	✗	84.9%
Enhanced TEVAD (Ours)	✓	✓	✓	85.3%

SwinBERT trained with UCA enhances anomaly detection accuracy, demonstrating the value of multimodal learning for optimizing surveillance tasks.

Video summarization

Method	Features	B1	B2	B3	B4	M	R	C
S2VT [42]	Inception V4	28.27	16.34	9.62	5.56	10.35	25.38	15.44
S2VT [42]	VGG 16	23.53	13.17	7.72	4.53	10.39	24.09	13.92
RecNet_global [43]	Inception V4	25.81	15.06	9.14	5.78	10.42	25.12	16.90
RecNet_local [43]	Inception V4	26.89	15.73	9.52	6.00	10.55	25.52	16.70
MARN [32]	Inception V4 ResNext101	26.16	15.61	10.22	6.63	9.67	23.85	15.33
SGN [35]	ResNet101 ResNext101	29.17	16.93	10.32	6.28	11.73	26.25	18.95
SwinBERT [22]	VidSwin	25.02	15.60	9.98	6.33	11.15	27.16	25.29
CoCap [37]	CLIP	28.53	17.34	10.75	6.57	11.43	28.14	21.24

Natural language video retrieval

Method	IoU=0.3		IoU=0.5		IoU=0.7	
	R@1	R@5	R@1	R@5	R@1	R@5
CTRL [12]	6.35	20.09	3.00	10.88	0.74	3.32
SCDM [49]	6.49	10.72	3.65	7.39	1.66	4.78
A2C [14]	4.25	-	1.78	-	0.44	-
2D-TAN [52]	7.76	19.67	4.16	11.54	2.22	6.00
LGI [30]	7.71	-	3.26	-	1.18	-
MMN [47]	8.68	21.38	4.66	11.98	2.12	5.89
MomentDiff [18]	8.15	-	4.46	-	1.71	-

Mainstream models perform far worse on surveillance videos than on general videos, highlighting their inherent ambiguity and semantic complexity.

Dense captioning

Method	Features	Recall				Precision			
		0.3	0.5	0.7	0.9	0.3	0.5	0.7	0.9
TDA-CG [44]	C3D	32.54	16.64	8.40	2.49	58.79	27.16	8.95	1.44
PDVC [45]	C3D	48.03	28.48	11.80	2.50	67.33	36.19	12.20	1.94
TDA-CG [44]	I3D	31.72	17.09	8.97	2.89	57.88	29.69	12.45	1.91
PDVC [45]	I3D	49.53	28.59	12.56	2.11	68.84	35.36	12.84	1.73

Yuan, et al. "Towards surveillance video-and-language understanding: New dataset baselines and challenges. **CVPR**. 2024.

4 Overview of VLM-based VAC

A Preliminary Exploration of VLM — Reasoning

Proble m

Current multimodal methods integrate visual and text features but lack cross-temporal anomaly modeling (e.g., instant explosions vs. prolonged theft), leading to the following issues:

Problem1: Lack of multi-granularity data

Annotation:Explosion



Coarse labels

[5.0, 10,2]The explosion happened suddenly



Local clip annotation

Single-granularity labels mismatch the multi-scale nature of real anomalies (instant actions / continuous behavior chains), preventing progressive reasoning.

Problem2: Limited fine-grained annotations



High annotation cost



Subjective bias

High-quality fine-grained annotations (e.g., UCA) require intensive manual effort, causing efficiency bottlenecks and subjective bias.

Problem3 : Weak interpretability

Why is it abnormal?



Insufficient anomaly judgment

What caused the exception?



Lack of causal analysis

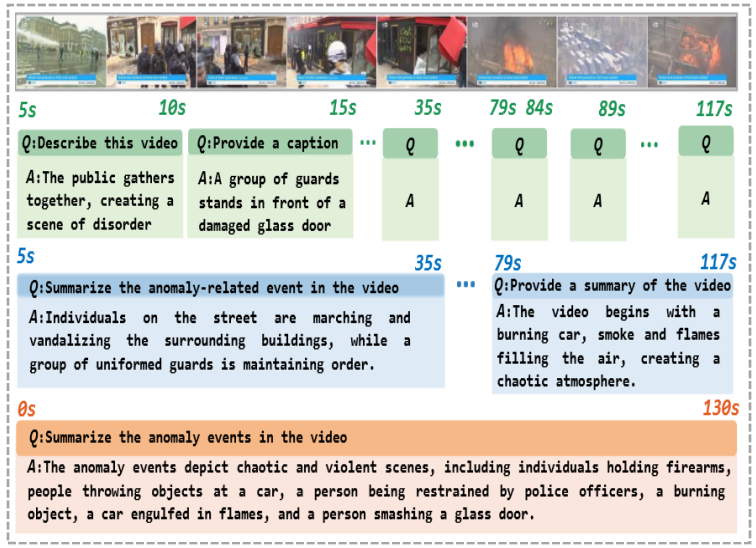
Models only learn shallow visual-text correspondences, failing to capture the core logic of anomaly understanding.

4 Overview of VLM-based VAC

A Preliminary Exploration of VLM — Reasoning

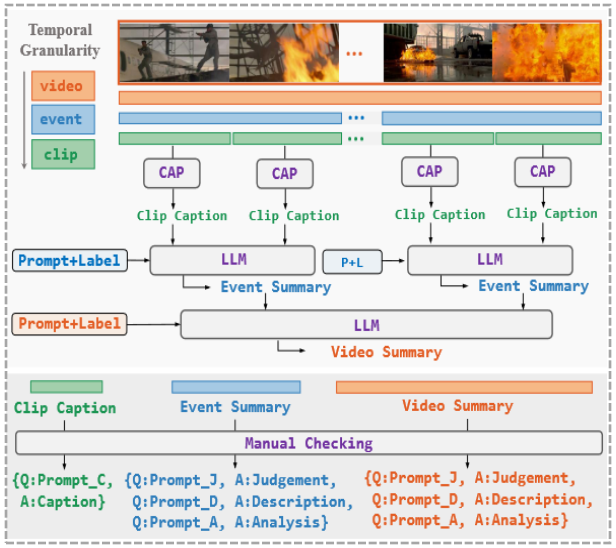
Thought Build a multi-granularity anomaly understanding benchmark based on large-model capabilities, focusing on hierarchical representation, automatic annotation, and structural reasoning.

Thought1: Build a three-level hierarchical annotation system



Establish a Clip–Event–Video anomaly representation framework covering the full process from instant perception to long-term reasoning.

Thought2: develop a semi-automatic LLM annotation engine



Combine manual key event boundary labeling with recursive text generation using VLM (LLaVA) and LLM (LLaMA3) for multi-granularity semantic extraction in a two-stage automated pipeline.

Thought3 : design multi-dimensional structured instruction templates

Judgement

1. "What types of anomalies are shown in the video clip?"
2. "Are there any anomaly events detected in the video?"
3. "Detect and classify the anomaly events in the video."
4. "Identify any abnormal behaviors depicted in the video."

Description

1. "Describe the anomaly events observed in the video."
2. "Could you describe the anomaly events observed in the video?"
3. "Could you specify the anomaly events present in the video?"
4. "Give a description of the detected anomaly events in this video."

Analysis

1. "Why do you judge this event to be anomalous?"
2. "Can you provide the reasons for considering it anomalous?"
3. "Can you give the basis for your judgment of this event as an anomaly?"
4. "What led you to classify this event as an anomaly?"

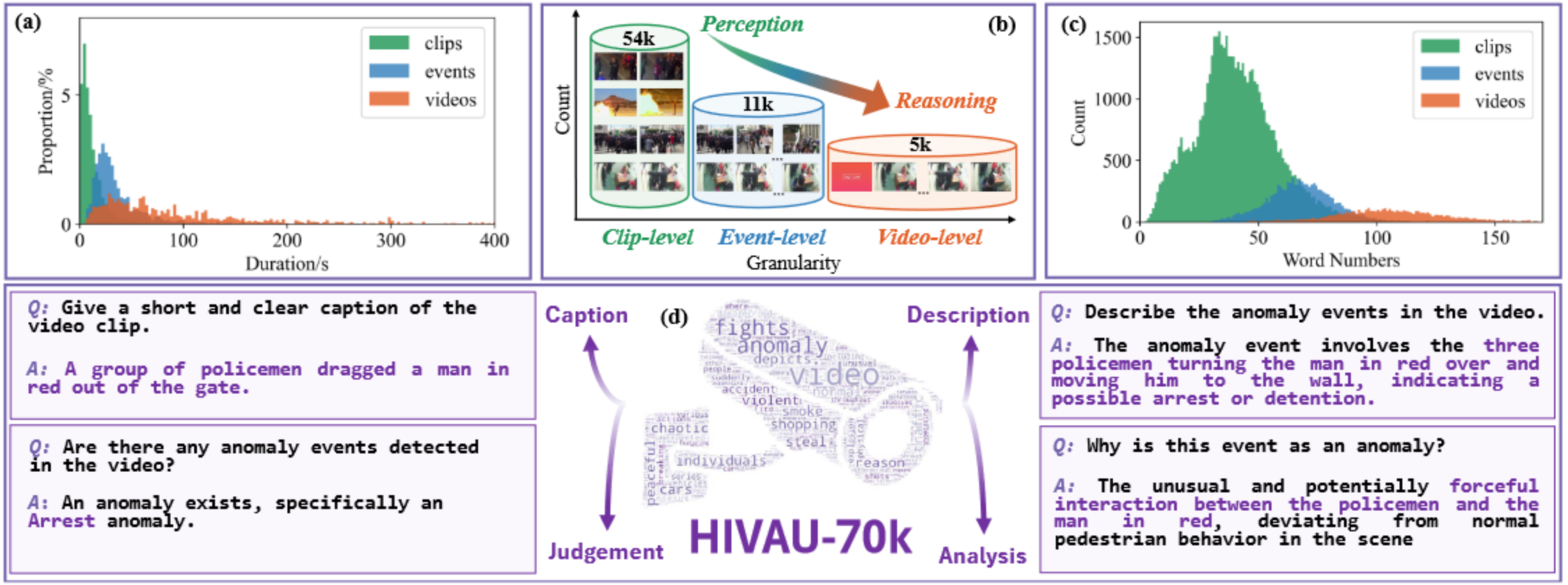
Design "Judgment–Description–Analysis" instruction templates to explicitly train anomaly reasoning ability.

4 Overview of VLM-based VAC

A Preliminary Exploration of VLM — Reasoning

Method

HIVAU-70k is the first large-scale benchmark dataset with three-level annotations: clip, event, and video, enabling models to learn progressively from instantaneous visual perception to long-term causal reasoning.



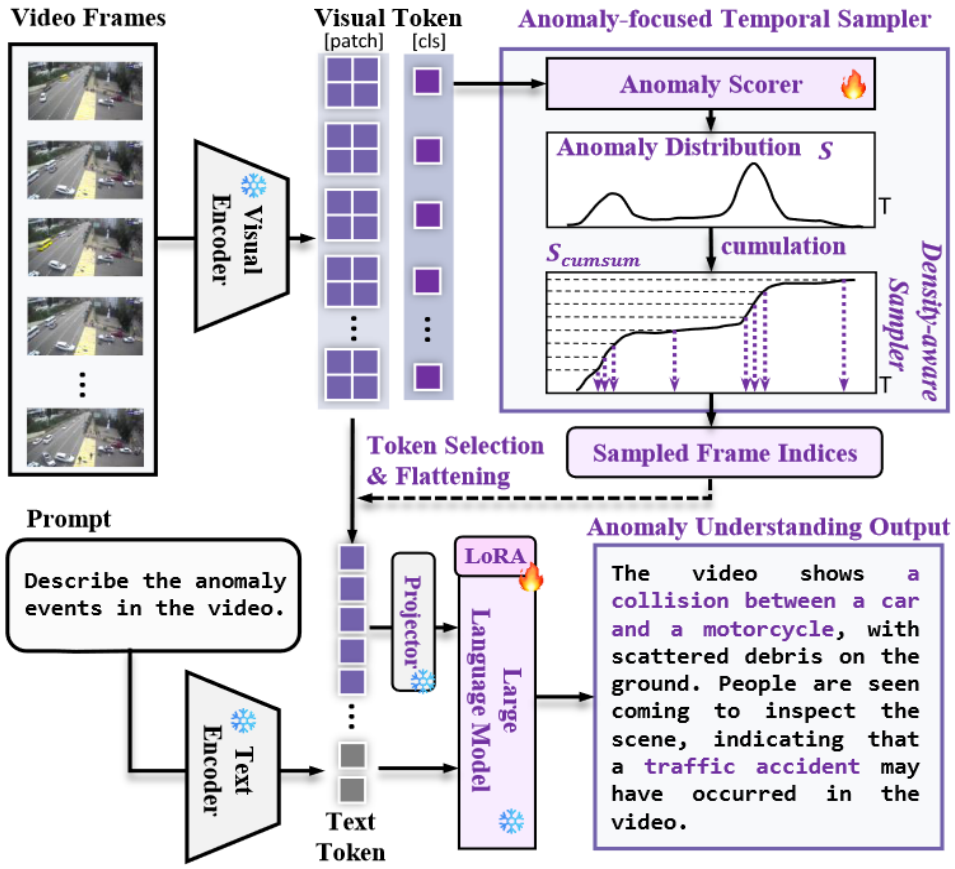
Zhang et al. Holmes-vau: Towards long-term video anomaly understanding at any granularity. **CVPR**. 2025.

4 Overview of VLM-based VAC

A Preliminary Exploration of VLM — Reasoning

Method

Holmes-VAU integrates an Anomaly-Focused Temporal Sampler (ATS) with multimodal large model fine-tuning, becoming the first long-video anomaly understanding framework to balance efficiency and interpretability.



LLM:InternVL2-2B

Zhang et al. Holmes-vau: Towards long-term video anomaly understanding at any granularity. **CVPR**. 2025.

4 Overview of VLM-based VAC

A Preliminary Exploration of VLM — Reasoning

Result

Using the three-level annotations of HIVAU-70k to provide cross-scale supervision, Holmes-VAU captures the spatiotemporal dependencies of anomalies and addresses the missed detections caused by traditional single-granularity labels.

Method	Params	BLEU [39](↑)			CIDEr [49](↑)			METEOR [4](↑)			ROUGE [26](↑)		
		C	E	V	C	E	V	C	E	V	C	E	V
Video-ChatGPT [37]	7B	0.152	0.068	0.066	0.033	0.011	0.013	0.102	0.069	0.044	0.153	0.048	0.079
Video-LLaMA [65]	7B	0.151	0.079	0.104	0.024	0.014	0.017	0.112	0.076	0.057	0.156	0.067	0.090
Video-LLaVA [25]	7B	0.164	0.046	0.055	0.032	0.009	0.013	0.097	0.022	0.014	0.132	0.023	0.045
LLaVA-Next-Video [68]	7B	0.435	0.091	0.120	0.102	0.015	0.031	0.117	0.085	0.096	0.198	0.080	0.106
QwenVL2 [51]	7B	0.312	0.082	0.155	0.044	0.020	0.044	0.133	0.092	0.112	0.163	0.081	0.137
InternVL2 [6]	8B	0.331	0.101	0.145	0.052	0.022	0.035	0.141	0.095	0.101	0.182	0.102	0.122
Holmes-VAU (Ours)	2B	**0.913**	**0.804**	**0.566**	**0.467**	**1.519**	**1.437**	**0.190**	**0.165**	**0.121**	**0.329**	**0.370**	**0.355**



Prompt: Determine whether there are anomaly events in the video and the specific name of the anomaly.

Ground Truth: The anomaly exists and is specifically identified as a Burglary event.

InternVL2-8B [Incorrect Judgement]
Yes, there are anomaly events in the video. The specific anomaly is that the man in the video is seen throwing a punch at the other man.

QwenVL2-7B [Incorrect Judgement]
There are no anomaly events in the video.

Holmes-VAU (Ours)
Yes, there is an anomaly in the video, specifically a break-in burglary.



Prompt: Identify any abnormal events depicted in the video.

Ground Truth: No anomaly exists in the video

InternVL2-8B [Incorrect Judgement]
The video shows a white truck parked on the side of the road, which is an abnormal behavior as it is not supposed to be parked on the road.

QwenVL2-7B
The video does not show any abnormal events.

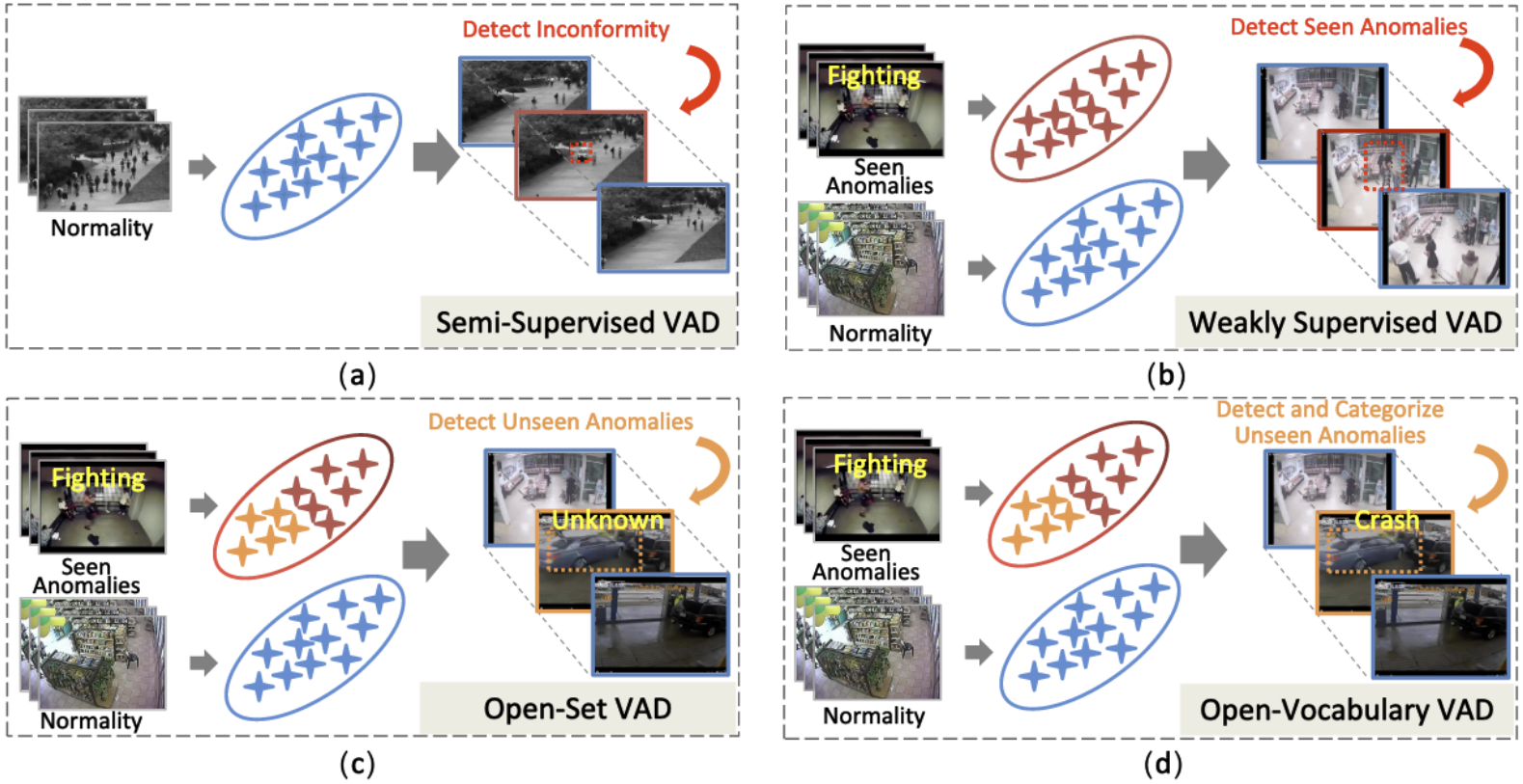
Holmes-VAU (Ours)
There is no anomaly in the video.

5 Overview of VLM-based OVAD

A Preliminary Exploration of VLM — Generalization

Problem

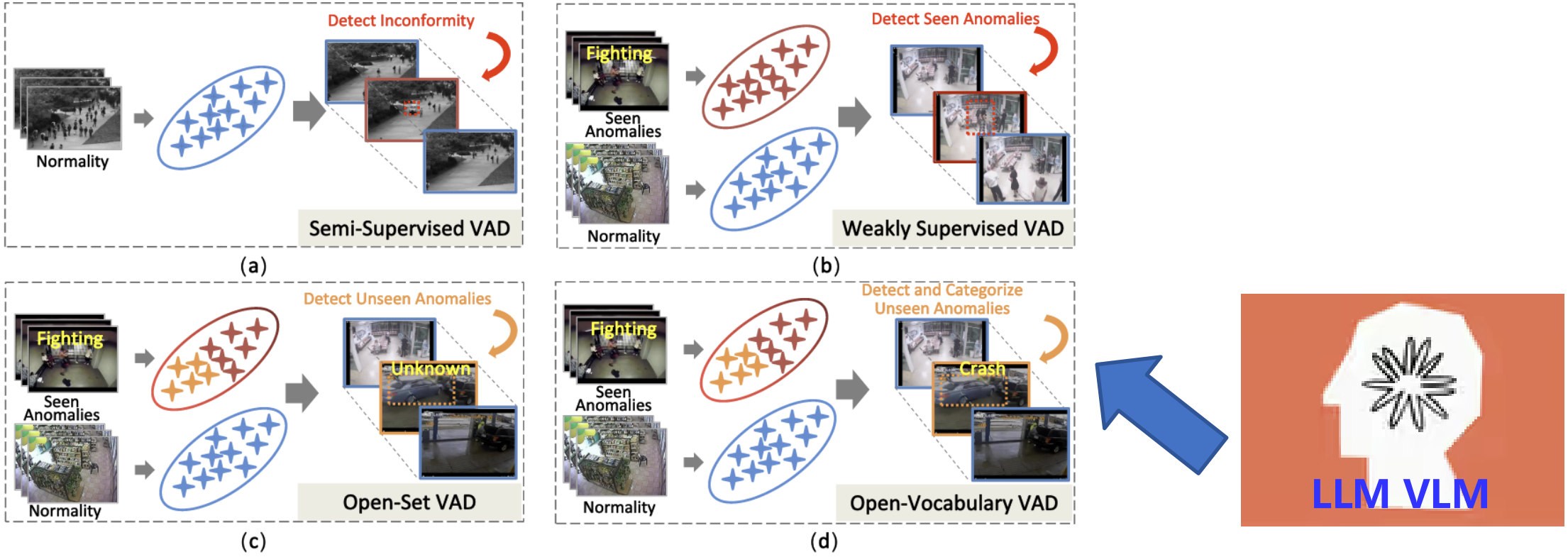
Current VAD methods are essentially confined to closed-set classification problems, making it difficult to detect unseen anomaly categories, which severely limits their applicability in real-world scenarios.



5 Overview of VLM-based OVAD

A Preliminary Exploration of VLM — Generalization

Thought Explore open-vocabulary anomaly detection using VLMs to leverage their general knowledge and generalization for open-world recognition.

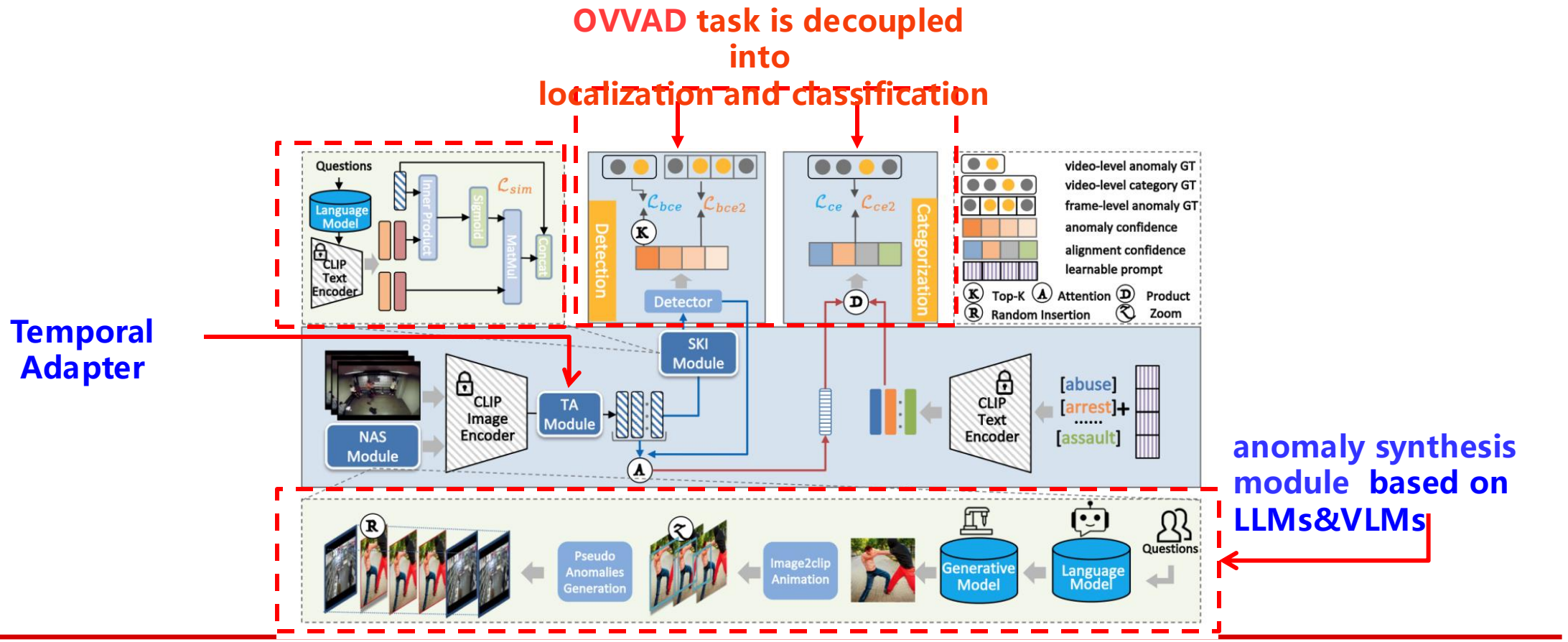


5 Overview of VLM-based OVAD

A Preliminary Exploration of VLM — Generalization

Method

Decouple the open-vocabulary VAD task and introduce a temporal adapter module, a semantic knowledge injection module, and an anomaly synthesis module.



Wu P, et al. Open-vocabulary video anomaly detection. **CVPR**, 2024.

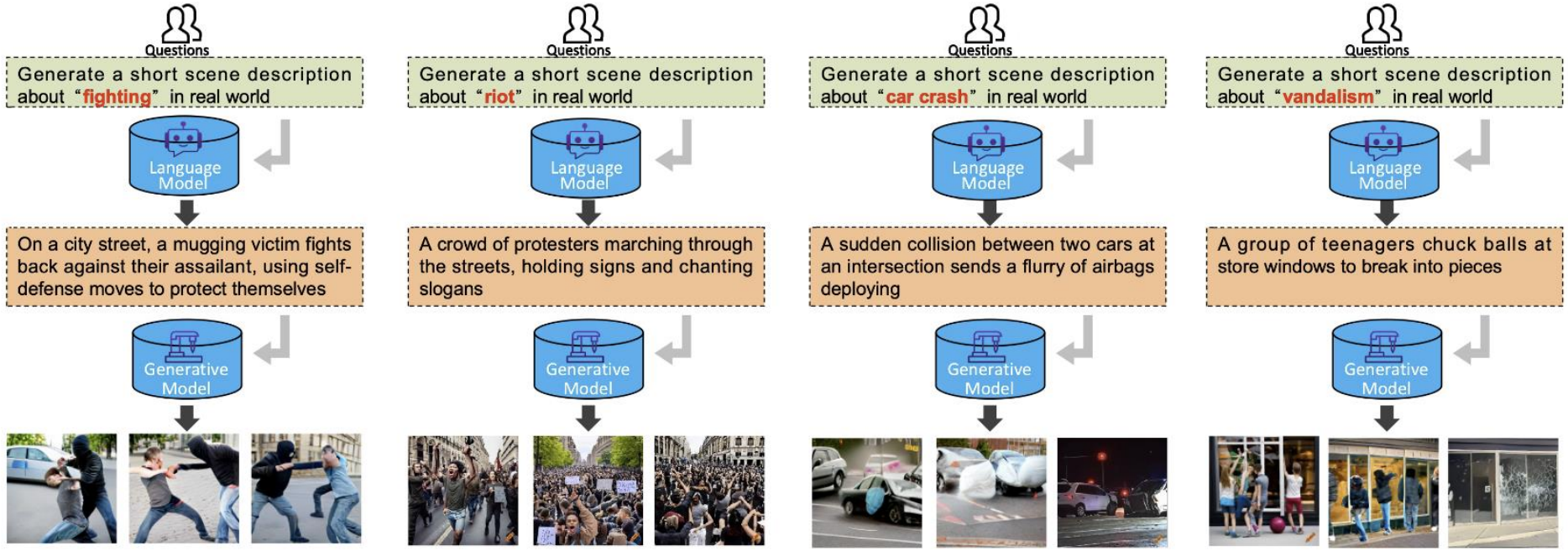
5 Overview of VLM-based OVAD

A Preliminary Exploration of VLM — Generalization

Method

Decouple the open-vocabulary VAD task and introduce a temporal adapter module, a semantic knowledge injection module, and an anomaly synthesis module.

anomaly synthesis module



anomaly synthesis:

Generate textual descriptions of anomalies, synthesize pseudo-images based on these descriptions, simulate pseudo-videos through animation, and randomly insert them into normal videos.

Wu P, et al. Open-vocabulary video anomaly detection. **CVPR**, 2024.

5 Overview of VLM-based OVAD

A Preliminary Exploration of VLM — Generalization

Result

As the first method to achieve open-vocabulary anomaly detection, it attains state-of-the-art performance in detecting unknown anomalies.

Mode	Method	AUC(%)	AUC_b(%)	AUC_n(%)
Semi	SVM baseline	50.10	N/A	N/A
	OCSVM[36]	63.20	N/A	N/A
	Hasan et al.[9]	51.20	N/A	N/A
Weak	Sultani et al.†[38]	84.14	N/A	N/A
	Wu et al.†[51]	84.57	N/A	N/A
	AVVD†[52]	82.45	N/A	N/A
	RTFM†[42]	85.66	N/A	N/A
	DMU†[62]	86.75	N/A	N/A
	UMIL†[23]	86.75	N/A	N/A
	CLIP-TSA†[12]	87.58	N/A	N/A
Weak	Zhu et al.*[67]	78.82	N/A	N/A
	Sultani et al.[38]	78.25	86.31	80.12
	Wu et al.[51]	82.24	90.62	84.13
	RTFM[42]	84.47	92.54	85.87
	DMU[62]	85.14	93.52	86.24
	Ours	**86.40**	**93.80**	**88.20**

Mode	Method	AP(%)	AP_b(%)	AP_n(%)
Semi	SVM baseline	50.80	N/A	N/A
	OCSVM[36]	28.63	N/A	N/A
	Hasan et al.[9]	31.25	N/A	N/A
Weak	Sultani et al.†[38]	75.18	N/A	N/A
	Wu et al.†[51]	80.00	N/A	N/A
	RTFM†[42]	78.27	N/A	N/A
	AVVD†[52]	78.10	N/A	N/A
	DMU†[62]	82.41	N/A	N/A
	CLIP-TSA†[12]	82.17	N/A	N/A
Weak	Zhu et al.*[67]	64.40	N/A	N/A
	Sultani et al.[38]	52.26	51.25	54.64
	Wu et al.[51]	55.43	52.94	64.10
	RTFM[42]	58.99	55.72	65.97
	DMU[62]	63.90	**60.12**	71.63
	Ours	**66.53**	57.10	**76.03**

For unknown anomaly detection, the proposed method achieves improvements of 2% and 4% on the UCF-Crime and XD-Violence datasets, respectively.

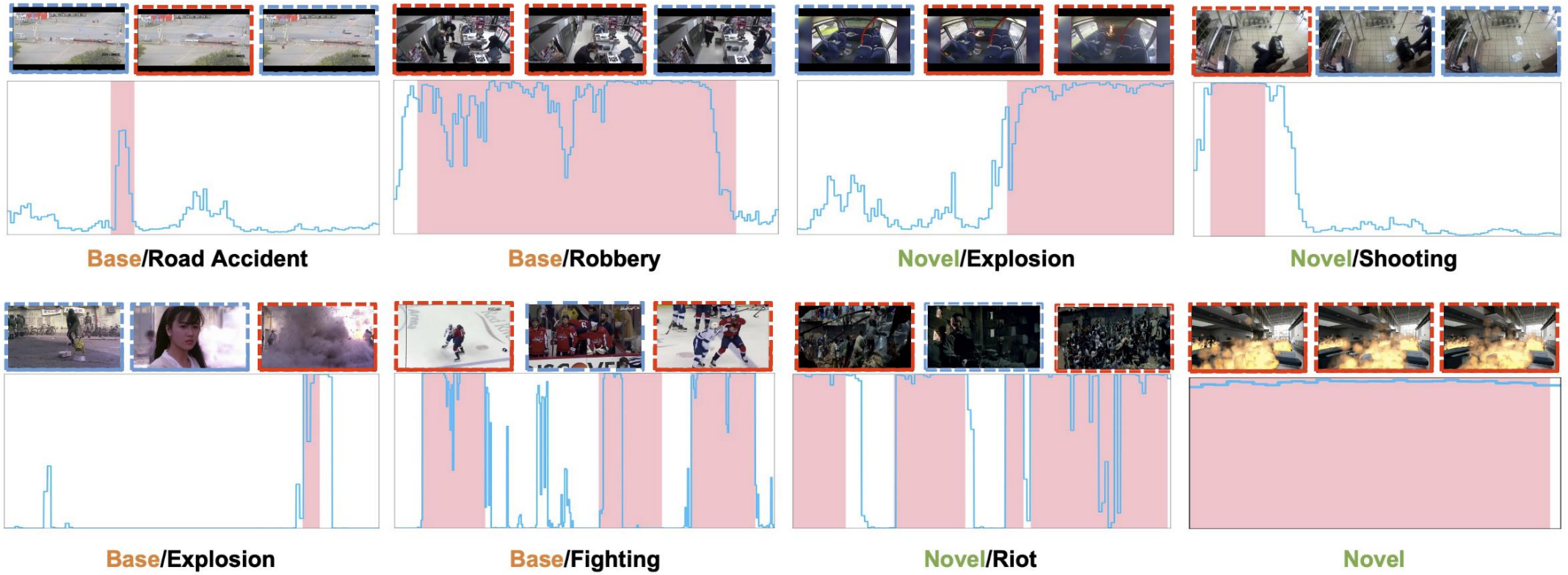
	ACC(%)	ACC_b(%)	ACC_n(%)
w/o NAS	37.86	43.14	34.83
Finetune_N	39.29	37.25	**40.45**
Finetune_N+B(Ours)	**41.43**	**49.02**	37.08
w/o NAS	59.60	**91.98**	15.18
Finetune_N	62.03	82.06	**34.55**
Finetune_N+B(Ours)	**64.68**	89.31	30.90

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Result

As the first method to achieve open-vocabulary anomaly detection, it attains state-of-the-art performance in detecting unknown anomalies.



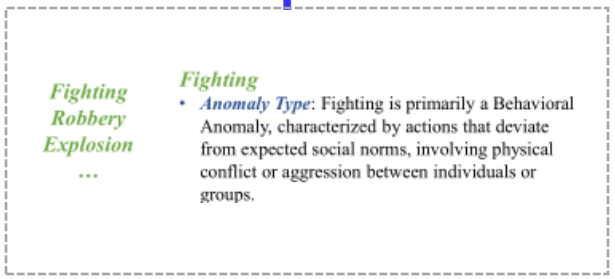
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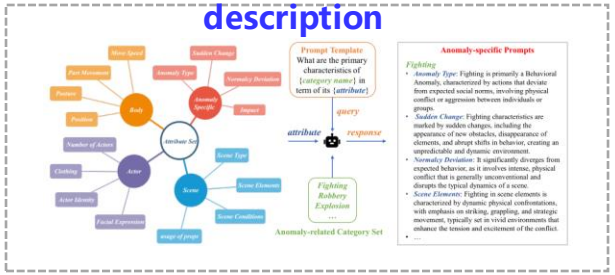
Problem

OVVAD method relies solely on the pre-trained CLIP model, generates concise textual descriptions, and does not leverage more general image-based VLMs.

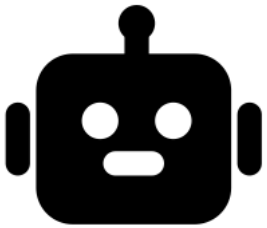
Issue: concise textual descriptions



Single label and single description



Diverse category- and attribute-level descriptions



LLM



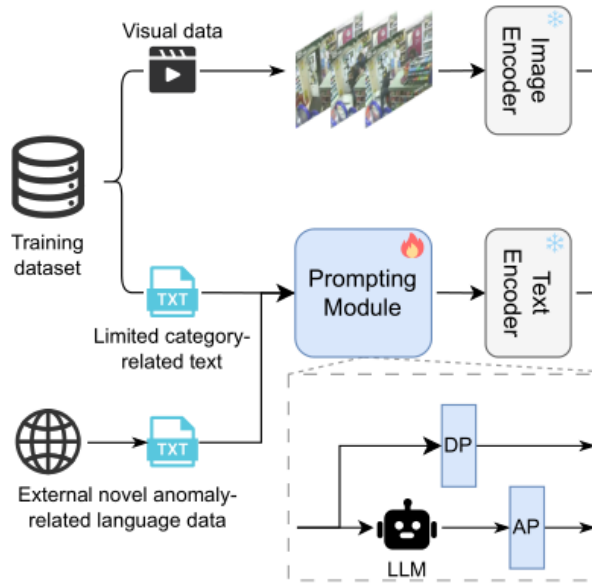
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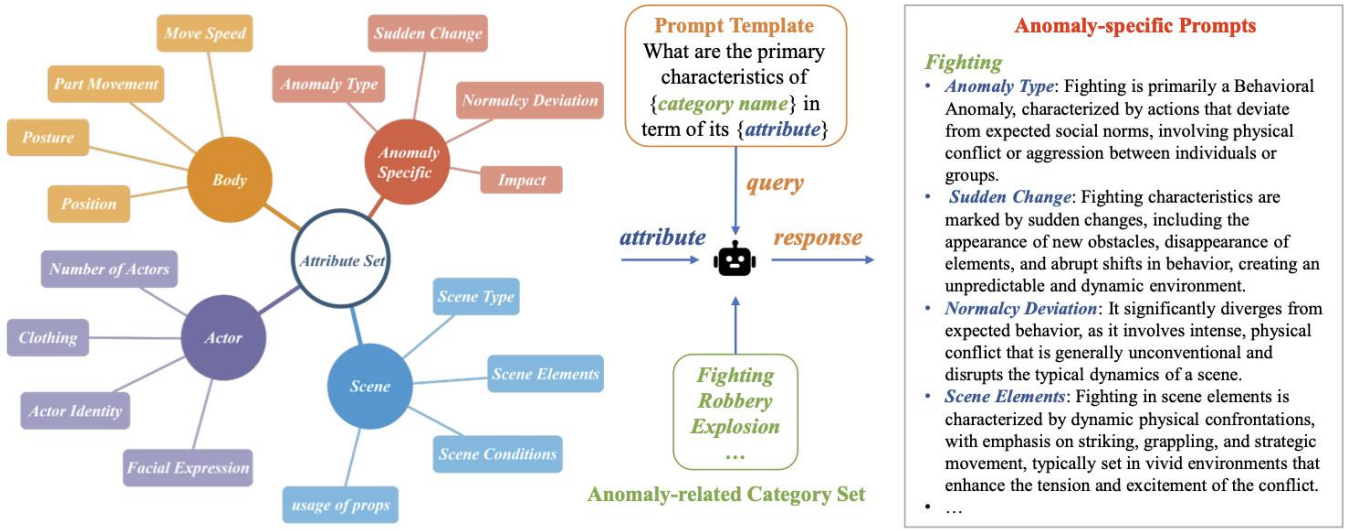
A Preliminary Exploration of VLM — Generalization

Thought

Leverage prompt tuning to mine the general knowledge within I-VLM for OVVAD, employing anomaly-specific prompts generated by large language models to enhance the generalization of the classification subtask.

Generate anomaly-specific prompts using I-VLM





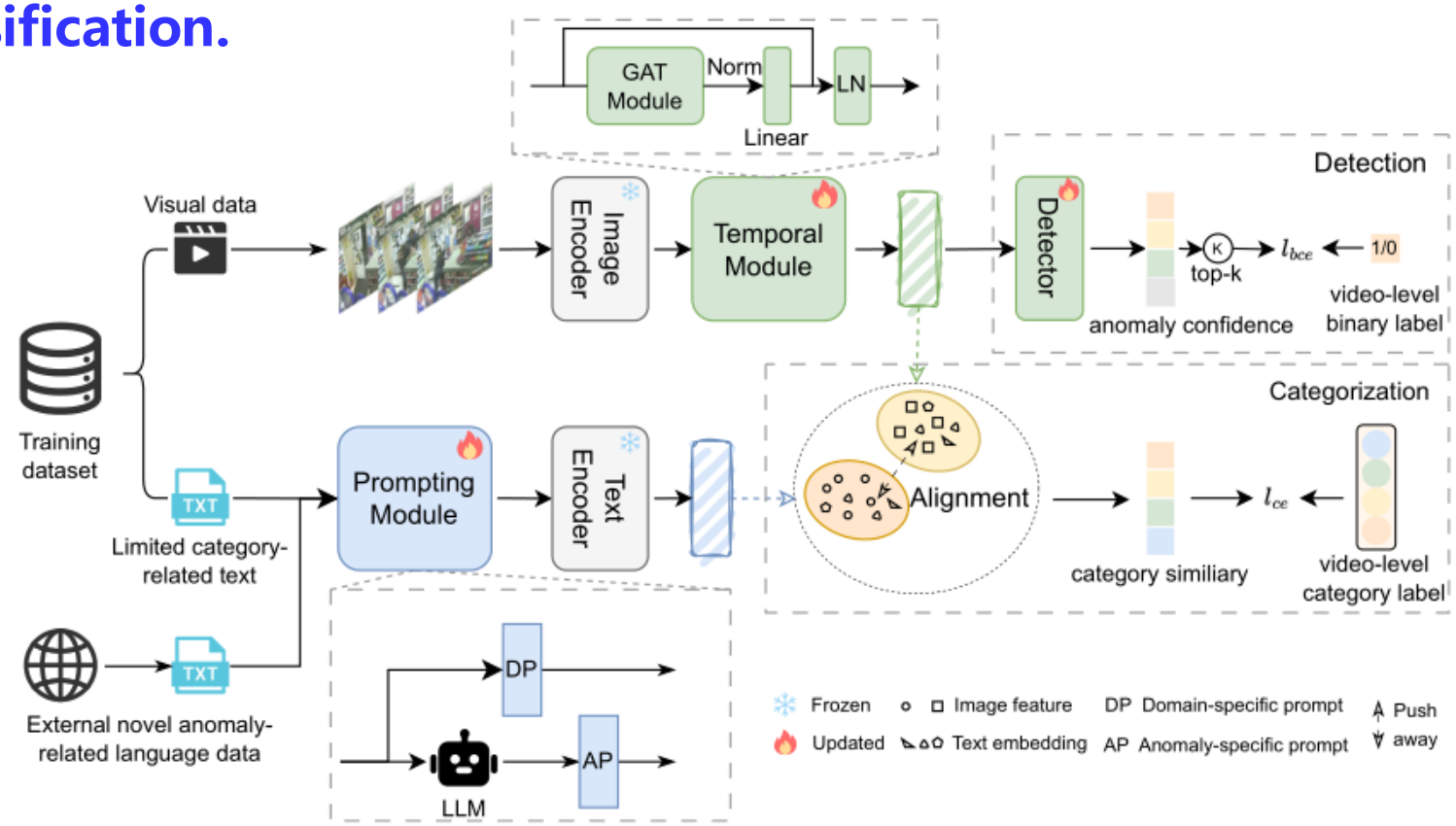
Use category + attribute + query templates to enable the large model to generate richer and more detailed textual content

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A Preliminary Exploration of VLM — Generalization

Method

PLOVAD introduces a prompt module based on I-VLM, combining domain-specific knowledge (prompt templates) with anomaly-specific knowledge (generated by LLMs) to jointly enhance the generalization ability of anomaly classification.



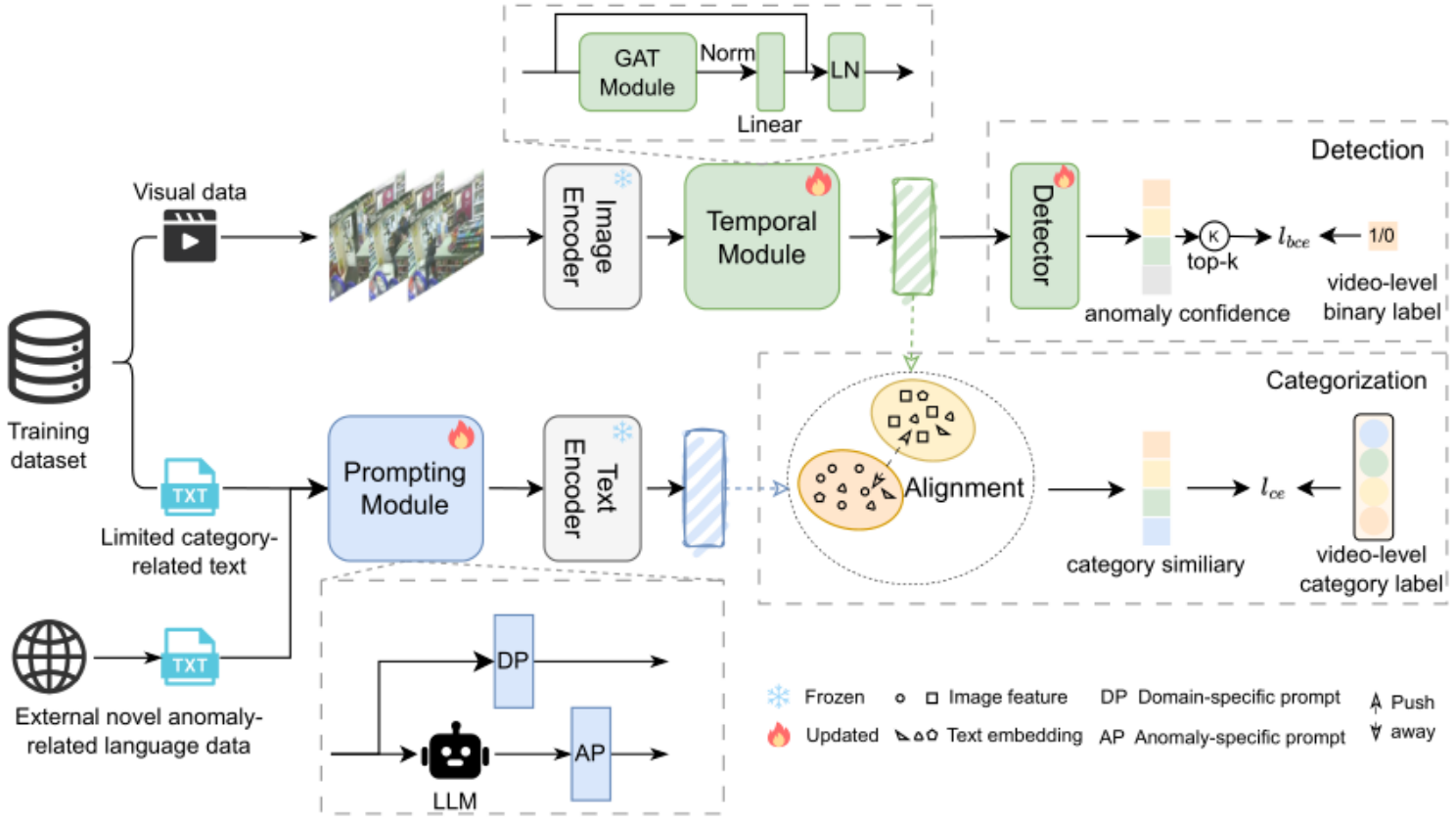
C. Xu, K. Xu, X. Jiang and T. Sun, PLOVAD: Prompting Vision-Language Models for Open Vocabulary Video Anomaly Detection. **TCSVT**, 2025.

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A Preliminary Exploration of VLM — Generalization

Method

PLOVAD introduces a temporal module utilizing a lightweight Graph Attention Network (GAT), which efficiently performs temporal modeling while reducing computational overhead.



C. Xu, K. Xu, X. Jiang and T. Sun, PLOVAD: Prompting Vision-Language Models for Open Vocabulary Video Anomaly Detection. **TCSVT**, 2025.

5 Overview of VLM-based OVAD

A Preliminary Exploration of VLM — Generalization

Result

PLOVAD with stronger generalization capability, outperforms previous approaches across multiple datasets.

TABLE I

DETECTION PERFORMANCE COMPARISONS(AUC(%)) UNDER DIFFERENT SETTINGS ON UCF-CRIME AND SHANGHAITECH. † MEANS THE METHOD IS RE-IMPLEMENTED WITH CLIP FEATURE.

Supervision	Method	Feature	UCF-Crime	ShanghaiTech
Semi-Supervised	Hasan (2016,CVPR) [5]	-	50.60	60.85
	BODS (2019, ICCV) [6]	I3D	68.26	-
	GODS (2019, ICCV) [6]	I3D	70.46	-
	HF^2VAD (2021, ICCV) [8]	I3D	-	76.20
	GCL (2022, CVPR) [7]	ResNext	74.20	79.62
zero-shot	CLIP (2021, ICML) [32]	CLIP	52.80	50.37
Weakly-Supervised	Sultani et al. (2018, CVPR) [4]	C3D	75.41	-
	GCN (2019, CVPR) [16]	TSN	82.12	84.44
	RTFM (2021, ICCV) [1]	I3D	84.30	97.21
	RTFM† (2021, ICCV) [1]	CLIP	85.02	97.24
	DAR (2022, TIFS) [22]	I3D	85.18	97.54
	S3R (2022, ECCV) [18]	I3D	85.99	97.48
	MSL (2023, AAAI) [17]	I3D	85.30	96.08
	Shi et al. (2023, TMM) [25]	I3D	85.79	97.48
	UMIL (2023, CVPR) [20]	X-CLIP	86.75	96.78
	UR-DMU (2023, AAAI) [19]	I3D	86.97	-
	UR-DMU† (2023, AAAI) [19]	CLIP	86.07	97.92
	Fan et al. (2024, TCSVT) [24]	I3D	86.19	-
	PLOVAD (ours)	CLIP	**87.06**	**98.08**
Open Vocabulary	Random Baseline	CLIP	55.65	50.46
	CLIP† (2021, ICML) [32]		52.80	51.92
	RTFM† (2021, ICCV) [1]		84.70	96.69
	UR-DMU† (2023, AAAI) [19]		84.81	97.54
	Wu et al. (2024, CVPR) [30]		86.40	-
	PLOVAD (ours)		**86.78**	**97.98**

TABLE III

DETECTION PERFORMANCE COMPARISONS IN THE OPEN VOCABULARY SETTING ON XD-VIOLENCE. $AP_b(\%)$ IS EVALUATED ONLY ON THE BASE SAMPLES AND $AP_n(\%)$ IS EVALUATED ONLY ON THE NOVEL SAMPLES. HM IS THE HARMONIC MEAN OF AP_b AND AP_n.

Method	Feature	AP_b	AP_n	HM
Random Baseline	CLIP	11.21	21.18	14.66
CLIP† (2021, ICML) [32]		53.94	61.10	57.30
RTFM† (2021, ICCV) [1]		53.43	67.74	59.74
UR-DMU† (2023, AAAI) [19]		**64.93**	66.02	65.47
Wu et al. (2024, CVPR) [30]		57.10	**76.03**	65.22
PLOVAD (ours)		64.70	75.55	**69.70**

TABLE IV

DETECTION PERFORMANCE COMPARISONS IN THE OPEN VOCABULARY SETTING ON UBNORMAL

Method	Feature	AUC
Random Baseline	CLIP	52.62
CLIP† (2021, ICML) [32]		50.00
RTFM† (2021, ICCV) [1]		61.40
UR-DMU† (2023, AAAI) [19]		62.85
Wu et al. (2024, CVPR) [30]		62.94
PLOVAD (ours)		**64.35**

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Problem

Open-vocabulary VAD faces challenges such as detection ambiguity and classification confusion when encountering novel anomalies, leading to suboptimal performance in detecting and classifying unseen anomalies.

Issue 1: detection ambiguity



It fails to produce accurate anomaly scores for unseen anomaly categories

Issue 2: classification confusion



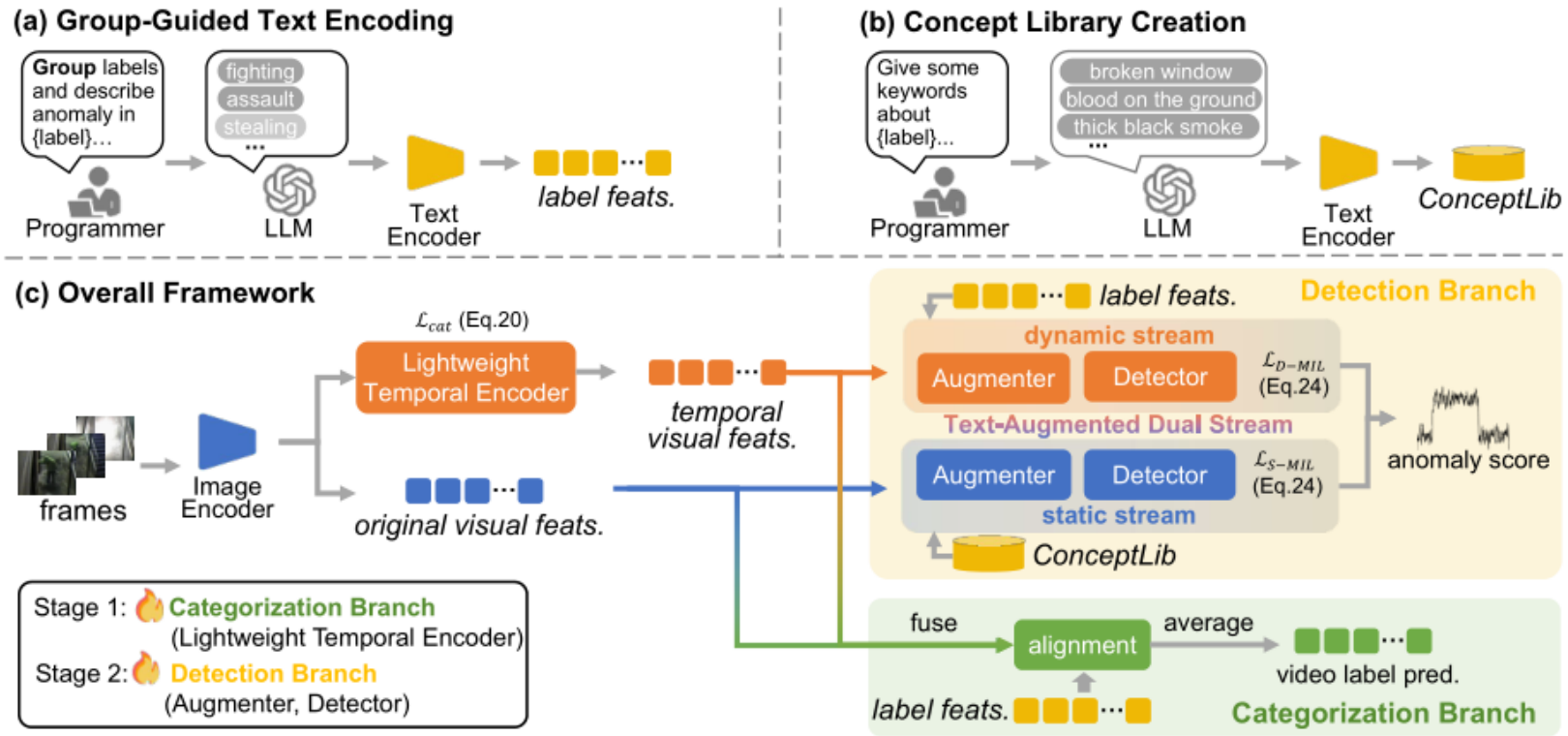
It tends to misclassify unseen anomaly categories as similar seen categories from the training set.

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A Preliminary Exploration of VLM — Generalization

Method

Anomize introduces a group-guided text encoding that leverages a VLM to cluster label categories and generate a descriptive text for each category group. It also constructs anomaly-related descriptions as a concept library.

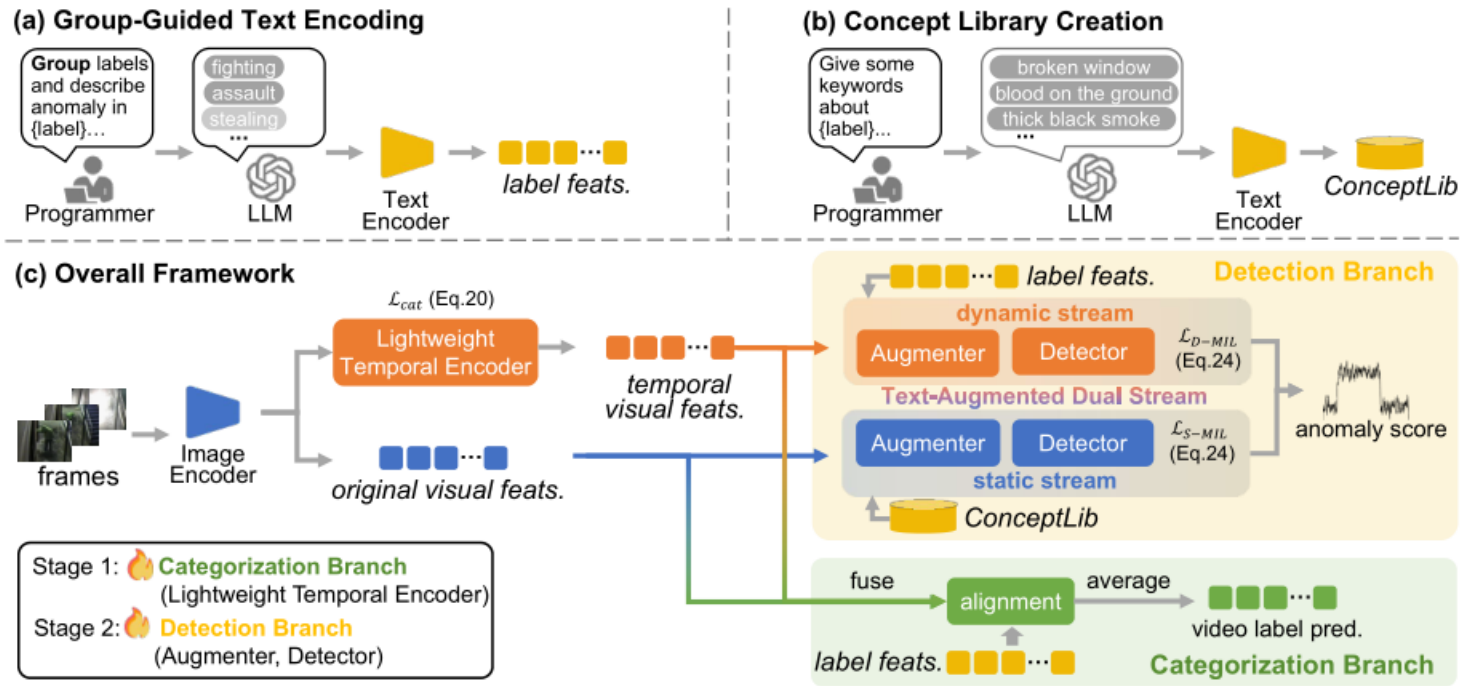


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A Preliminary Exploration of VLM — Generalization

Method

Anomize further introduces a text-augmented dual stream, where class labels are used for dynamic anomaly detection and the concept library is employed for static anomaly detection.



Li, Fei, et al. Anomize: Better Open Vocabulary Video Anomaly Detection. CVPR, 2025.

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A Preliminary Exploration of VLM — Generalization

Result

Anomize achieves leading performance across multiple open-vocabulary detection metrics.

Method	AP	AP_b	AP_n	AUC	AUC_b	AUC_n
Zhu et al.∗ [54]	64.40	-	-	78.82	-	-
Sultani et al. [33]	52.26	51.25	54.64	78.25	86.31	80.12
Wu et al. [41]	55.43	52.94	64.10	82.24	90.62	84.13
RTFM [34]	58.99	55.72	65.97	84.47	92.54	85.87
Wu et al. [42]	66.53	57.10	76.03	**86.40**	**93.80**	**88.20**
Ours	**69.31**	**57.37**	**84.24**	84.49	93.00	87.05

Table 1. **Detection Metrics (%) Comparisons for XD-VIOLENCE (left) and UCF-CRIME (right).** The best results are highlighted in bold, our method is shaded in gray, the symbol ∗ indicates different category divisions, and underlined values represent the second-best results.

Method	ACC	ACC_b	ACC_n	ACC	ACC_b	ACC_n
Wu et al. [42]	64.68	89.31	30.90	41.43	49.02	37.08
Ours	**90.29**	**92.37**	**87.43**	**47.14**	**56.86**	**41.57**

Table 2. **Top-1 Accuracy (%) Comparisons on XD-VIOLENCE (left) and UCF-CRIME (right).**

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OCT 19-23, 2025

Thanks!

Peng Wu
xdwupeng@gmail.com